

FMCG ERP

Supply Chain Module Manual

Version: 1.0

Date: 5 June 2026

Scope: Procurement · Inventory · Warehouses · WMS · Logistics

Chapters: 12

Field names and workflows reflect the actual ERP codebase. Screens may change as the ERP evolves.

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Supply Chain Module — Overview

Manual: FMCG ERP Supply Chain Module Manual

Audience: Procurement Officers, Warehouse Managers, Logistics Coordinators, Inventory Controllers

Scope: All supply chain screens, workflows, and import formats

What This Module Covers

Subsystem	Route	Purpose
Procurement	<code>/dashboard/procurement</code>	Purchase requisitions, orders, RFQ, suppliers, subcontracting
Inventory	<code>/dashboard/inventory</code>	Stock ledger, entries, issues, transfers, adjustments
Warehouses	<code>/dashboard/warehouses</code>	Warehouse master data, WMS zones and locations
WMS	<code>/dashboard/wms</code> (embedded in Warehouses)	Zones, locations, handling units, pick waves, quarantine
Logistics	<code>/dashboard/logistics</code>	International shipments, containers, customs clearance, fleet

Navigation

All Supply Chain modules are accessible from the left sidebar. Tab-level access is controlled by permissions.

Quick paths:

- New PR: **Sidebar** → **Procurement** → **Purchase Requests tab** → **New PR**
- Receive stock: **Sidebar** → **Inventory** → **Stock Entry (IN) tab**
- New warehouse: **Sidebar** → **Warehouses** → **Add Warehouse**
- Quarantine lot: **Sidebar** → **Warehouses** → **WMS/Putaway tab** → **Quarantine tab** → **Quarantine Stock**
- Track shipment: **Sidebar** → **Logistics** → **Overview tab**

Permission Map

Permission	Grants Access To
<code>procurement.view</code>	All procurement tabs (PRs, POs, RFQ, deliveries, etc.)
<code>inventory.view</code>	Stock ledger, movements, cycle count, shelf life, traceability
<code>inventory.adjust</code>	Adjust stock quantity for warehouses in scope
<code>inventory.adjust_all</code>	Adjust stock across all warehouses
<code>inventory.receive_all</code>	Receive stock into any warehouse
<code>inventory.dispatch_all</code>	Issue stock from any warehouse
<code>inventory.transfer_all</code>	Transfer stock between any warehouses
<code>inventory.delete</code>	Delete zero-quantity stock records in scope
<code>inventory.delete_all</code>	Delete zero-quantity stock records across all warehouses
<code>logistics.view</code>	All logistics tabs

Warehouse-scoped permissions use `canPerformInScope("warehouse", warehouse_id, action)` — a user with `inventory.adjust` (without `_all`) can only adjust stock in warehouses they are scoped to.

Chapters in This Manual

1. [Purchase Requisitions](#) — PR creation, line items, status workflow
2. [Purchase Orders](#) — PO fields, PR conversion, status, GRN
3. [RFQ & Supplier Quotations](#) — Request for quotation, responses, comparison
4. [Deliveries & Goods Receipt](#) — GRN creation, receipt against PO
5. [Suppliers](#) — Supplier master data, portal
6. [Blanket Agreements & Reorder](#) — Blanket PAs, reorder policies, suggestions
7. [Inventory Stock Ledger](#) — Stock summary, entry, issue, transfer, adjust
8. [Inventory Movements & Traceability](#) — Movement ledger, traceability, serials, valuation
9. [Warehouses](#) — Warehouse types, create, manage
10. [WMS — Zones, Locations & Operations](#) — Zones, bins, handling units, pick waves, quarantine
11. [Logistics & International Shipments](#) — Shipments, containers, customs, arrivals, fleet

Purchase Requisitions

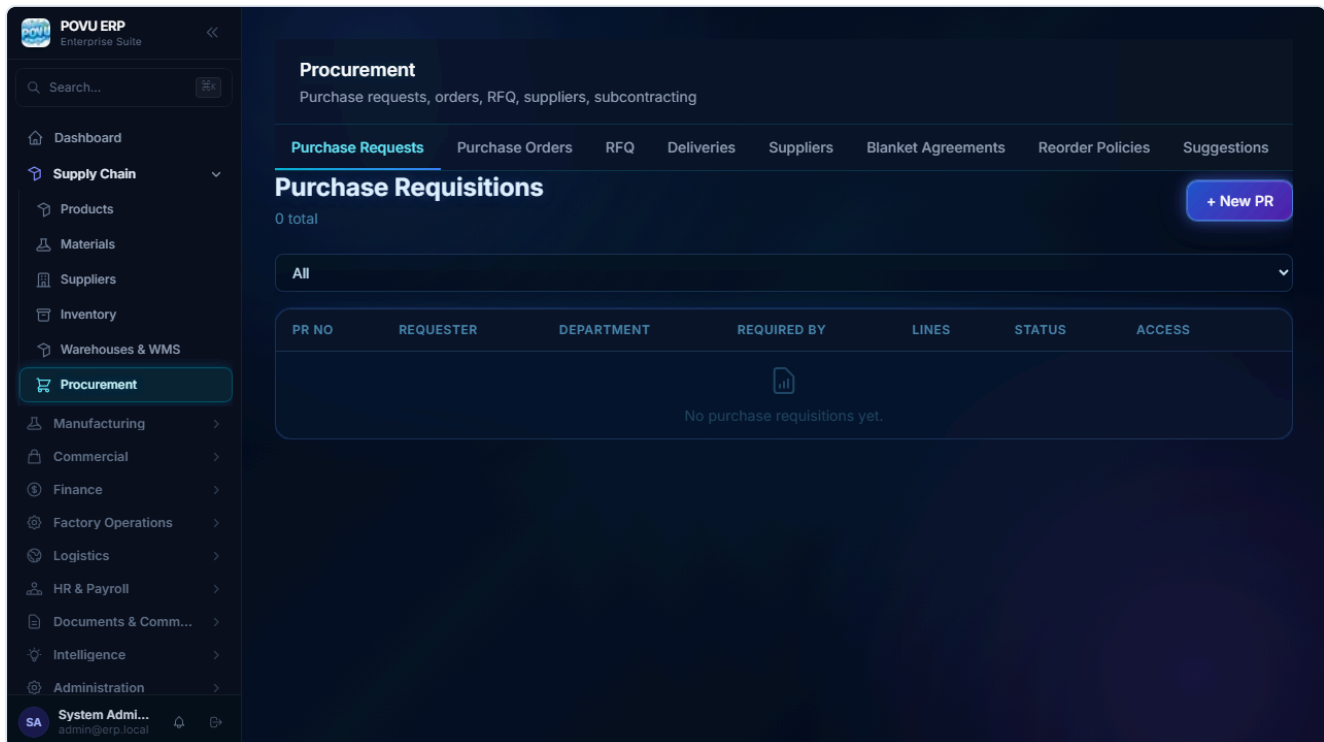
Route: `/dashboard/procurement` (default tab: Purchase Requests)

Permission required: `procurement.view`

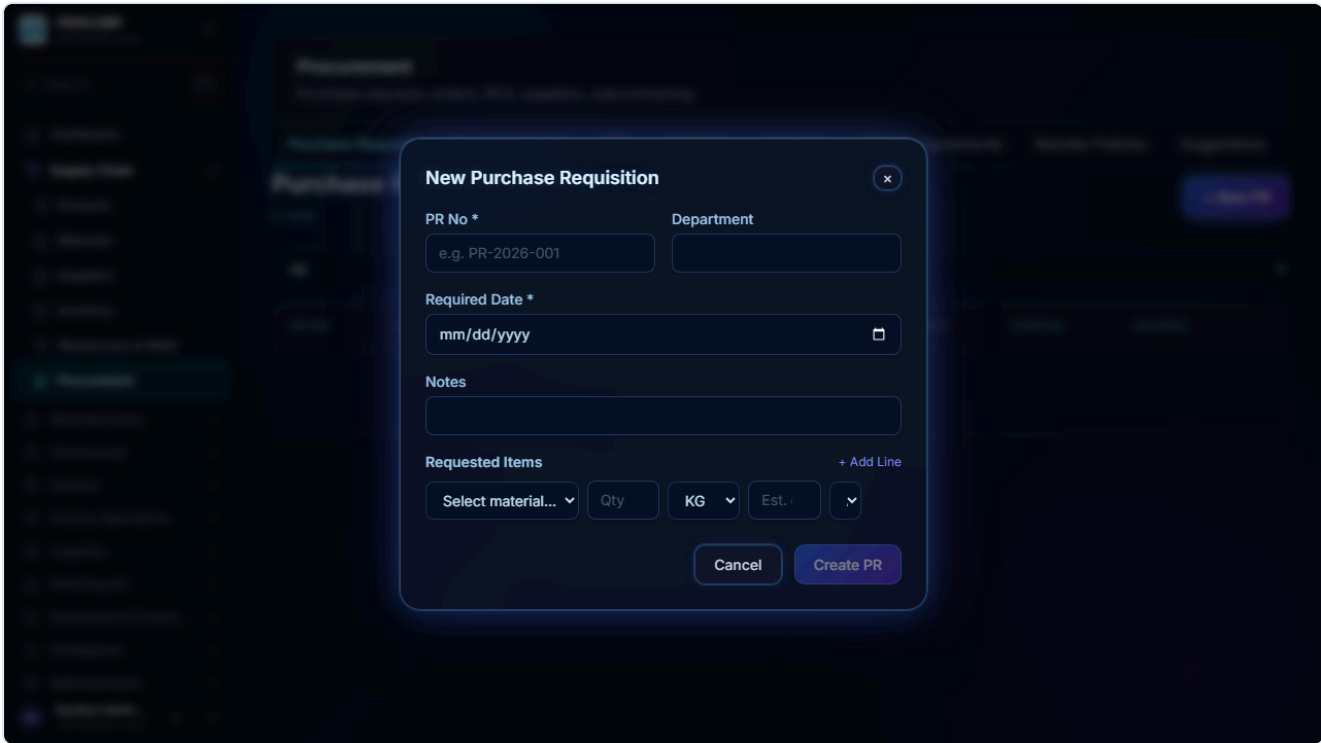
Tab key: `purchase-requests`

What It Does

A Purchase Requisition (PR) is an internal request to procure materials or products. PRs are raised by any department and go through an approval workflow before being converted into Purchase Orders. The system enforces scope-based access — users may be restricted to PRs within their company, branch, cost centre, or department.



Purchase Requests tab showing PR list with status badges and approval controls.



New PR form with line items, department, required date, and preferred supplier fields.

PR List

Filter

Status dropdown filters the list:

Filter Option	Value
All	(no filter)
Draft	DRAFT
Pending Approval	PENDING_APPROVAL
Approved	APPROVED
Converted to PO	CONVERTED
Rejected	REJECTED
Cancelled	CANCELLED

Table Columns

Column	Field	Notes
PR No	<code>pr_no</code>	Monospace; clickable link to PR detail at <code>/dashboard/procurement/{id}</code>
Requester	<code>requester_name</code>	Auto-populated from logged-in user
Department	<code>department</code>	Free text department name
Required By	<code>required_date</code>	Locale date format
Lines	<code>line_count</code>	Number of material lines on the PR
Status	<code>status</code>	Badge (APPROVED/CONVERTED = green; PENDING_APPROVAL = blue; REJECTED/CANCELLED = red; DRAFT = blue)
Access	<code>access.view_only</code>	"Actionable" (green) or "View only" (grey) badge based on scope permissions

Creating a Purchase Requisition

Button: + New PR (`data-testid="procurement-create-pr-button"`)

PR Header Fields

Field	Label	Required	Backend field	Notes
<code>pr_no</code>	PR No	Yes	<code>pr_no</code>	User-assigned; e.g. <code>PR-2026-001</code>
<code>department</code>	Department	No	<code>department</code>	Free text; e.g. "Production", "R&D"
<code>required_date</code>	Required Date	Yes	<code>required_date</code>	Date picker; the date materials are needed by
<code>notes</code>	Notes	No	<code>notes</code>	Free text; visible on PO conversion

Submit disabled until `pr_no` and `required_date` are filled.

PR Line Fields

Each PR can have one or more lines. Lines are added with the **+ Add Line** button. Minimum one line required (lines without `material_id` and `quantity` are excluded from submission).

Field	Label	Required	Backend field	Notes
<code>material_id</code>	Material	Yes*	<code>material_id</code>	Select from materials master (<code>code</code> – <code>name</code> format)
<code>quantity</code>	Qty	Yes*	<code>quantity</code>	Decimal; step 0.001
<code>unit</code>	Unit	Yes	<code>unit</code>	Default <code>KG</code> ; options: <code>KG / G / L / ML / PCS / BOX / CARTON / PALLET</code>
<code>estimated_unit_cost</code>	Est. cost	No	<code>estimated_unit_cost</code>	Decimal; step 0.01
<code>preferred_supplier_id</code>	Supplier	No	<code>preferred_supplier_id</code>	Optional preferred supplier for this line
<code>notes</code>	(no label)	No	<code>notes</code>	Per-line notes

*Lines without `material_id` + `quantity` are silently excluded from the submitted payload.

The backend also accepts `product_id` and `description` per line (from `PRLineCreate` schema), but these are not exposed in the current frontend create form.

PR Status Values

Status	Meaning	Badge Colour
<code>DRAFT</code>	Created; not submitted for approval	Blue
<code>PENDING_APPROVAL</code>	Submitted; awaiting approver action	Blue
<code>APPROVED</code>	Approved by authorised approver	Green
<code>CONVERTED</code>	PR lines converted to a Purchase Order	Green
<code>REJECTED</code>	Rejected by approver; <code>rejection_reason</code> populated	Red
<code>CANCELLED</code>	Cancelled by requester or admin	Red

PR Approval Workflow

DRAFT → PENDING_APPROVAL → APPROVED → CONVERTED
 ↘ REJECTED

Actions available from the PR detail page (`/dashboard/procurement/{id}`):

- **Submit for Approval** (DRAFT → PENDING_APPROVAL)
- **Approve** (PENDING_APPROVAL → APPROVED) — records `approved_by_id` and `approved_at`
- **Reject** (PENDING_APPROVAL → REJECTED) — requires `rejection_reason`
- **Convert to PO** (APPROVED → CONVERTED) — opens the Convert to PO flow

Convert PR to PO

Converting a PR to a Purchase Order requires:

Field	Backend field	Required	Notes
PO No	<code>po_no</code>	Yes	User-assigned PO reference
Supplier	<code>supplier_id</code>	Yes	Must select a supplier for all lines
Order Date	<code>order_date</code>	Yes	Date of the PO
Expected Delivery	<code>expected_delivery_date</code>	Yes	Date supplier is expected to deliver
Payment Terms	<code>payment_terms</code>	No	Free text; e.g. "Net 30"
Currency	<code>currency</code>	No	Default <code>USD</code>
Exchange Rate	<code>exchange_rate</code>	No	Default <code>1.0</code>
Notes	<code>notes</code>	No	Free text
Line Prices	<code>line_prices</code>	No	Override unit price per PR line: <code>{pr_line_id: unit_price}</code>

Scope-Based Access

PRs support multi-company, branch, and cost-centre scoping via `ProcurementScopeFields`:

Field	Purpose
<code>company_id</code>	Restricts PR to a specific legal entity

Field	Purpose
<code>branch_id</code>	Restricts PR to a branch/site
<code>cost_center_id</code>	Links PR to a cost centre for budget control
<code>department</code>	Free text department label

Users with limited scope see the "View only" badge on PRs outside their scope — they can read the record but cannot approve, edit, or convert it.

API Endpoints (Reference)

Method	Path	Description
GET	<code>/api/v1/procurement/purchase-requests/</code>	List PRs (filter: <code>status</code>)
POST	<code>/api/v1/procurement/purchase-requests/</code>	Create PR with lines
GET	<code>/api/v1/procurement/purchase-requests/{id}</code>	PR detail with lines
PATCH	<code>/api/v1/procurement/purchase-requests/{id}</code>	Update PR header (DRAFT only)
POST	<code>/api/v1/procurement/purchase-requests/{id}/approve</code>	Approve or reject
POST	<code>/api/v1/procurement/purchase-requests/{id}/convert-to-po</code>	Convert to PO

Purchase Orders

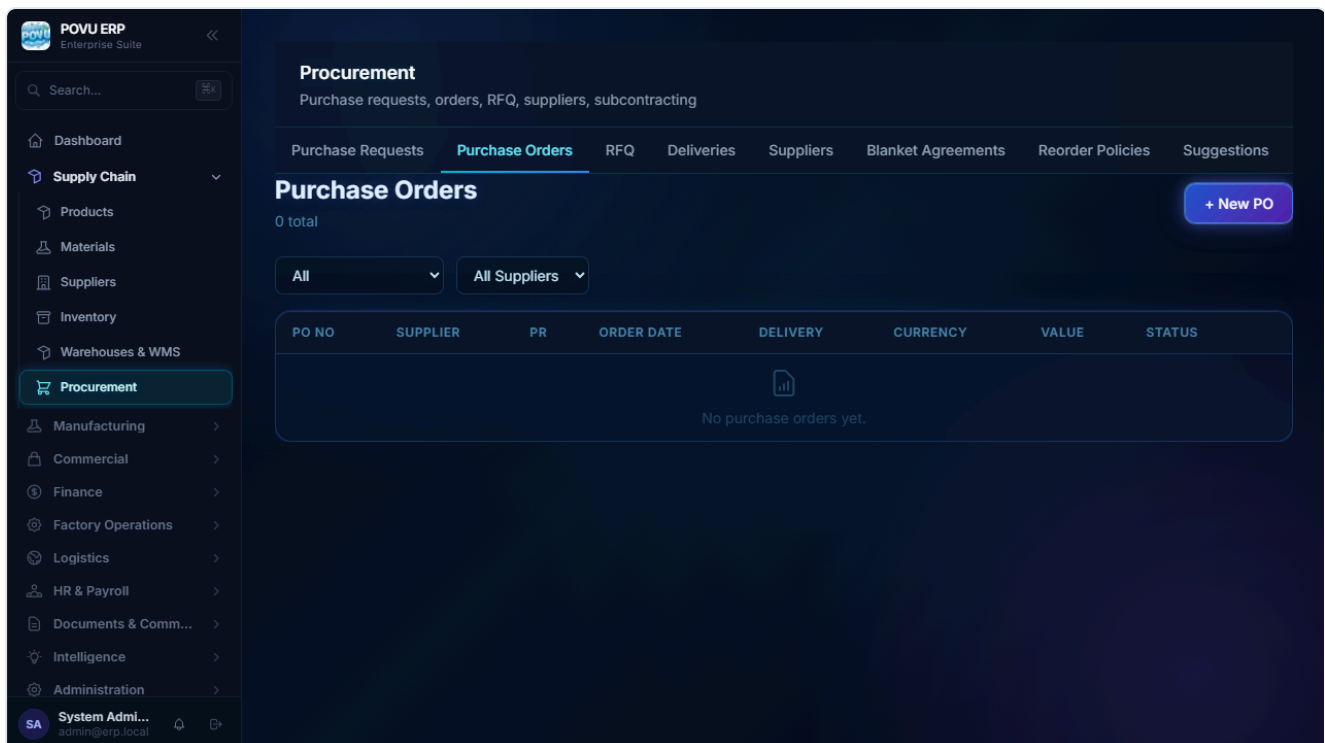
Route: `/dashboard/procurement?tab=orders`

Tab key: `orders`

Permission required: `procurement.view`

What It Does

Purchase Orders (POs) are the formal external documents sent to suppliers to procure materials or products. POs can be created by converting an approved PR, or created directly. Each PO tracks receipt progress through Goods Receipt Notes (GRNs).



Purchase Orders tab showing PO list with status, supplier, and receipt progress.

New Purchase Order

PO No * Currency

Supplier *

Order Date * Expected Delivery *

Payment Terms

Order Lines Qty [+ Add Line](#)

New Purchase Order modal — PO No, Supplier, Order Date, Delivery Date, and line items.

PO Header Fields

From the `POCreate` schema:

Field	Label	Required	Default	Notes
<code>po_no</code>	PO No	Yes	—	User-assigned; e.g. <code>PO-2026-001</code>
<code>supplier_id</code>	Supplier	Yes	—	Select from suppliers master
<code>order_date</code>	Order Date	Yes	—	Date PO is issued
<code>expected_delivery_date</code>	Expected Delivery	Yes	—	Date supplier should deliver
<code>payment_terms</code>	Payment Terms	No	—	Free text; e.g. "Net 30", "50% advance"
<code>currency</code>	Currency	No	<code>USD</code>	ISO currency code
<code>exchange_rate</code>	Exchange Rate	No	<code>1.0</code>	To KES; used for landed cost
<code>notes</code>	Notes	No	—	Internal notes
<code>lines</code>	Lines	No	<code>[]</code>	See PO Line Fields below

PO Line Fields

From `POLineCreate`:

Field	Backend field	Required	Default	Notes
<code>line_no</code>	Line No	Yes	—	Sequence
<code>material_id</code>	Material	No	—	Either <code>material_id</code> or <code>product_id</code>
<code>product_id</code>	Product	No	—	Either <code>material_id</code> or <code>product_id</code>
<code>pr_line_id</code>	PR Line	No	—	Links back to originating PR line
<code>description</code>	Description	No	—	Free text override
<code>ordered_quantity</code>	Ordered Qty	Yes	—	Decimal
<code>unit</code>	Unit	Yes	KG	UOM
<code>unit_price</code>	Unit Price	Yes	0	Decimal
<code>tax_rate</code>	Tax Rate	Yes	0	Decimal; e.g. 0.16 for 16% VAT

Computed fields on `POLineRead`:

- `line_total = ordered_quantity × unit_price × (1 + tax_rate)`
- `pending_quantity = max(ordered_quantity - received_quantity, 0)` — quantity not yet received via GRN

PO Status Values

Status	Meaning
DRAFT	PO created; not yet sent to supplier
SENT	PO dispatched to supplier
ACKNOWLEDGED	Supplier confirmed receipt of PO
PARTIALLY_RECEIVED	Some lines received via GRN; some pending
FULLY_RECEIVED	All lines fully received
CANCELLED	PO cancelled
CLOSED	PO administratively closed

PO Payment Status Values

Status	Meaning
UNPAID	Invoice received but payment not made
PARTIALLY_PAID	Part payment made
PAID	Fully paid
OVERPAID	Payment exceeds PO value

PO Detail Page

Route: `/dashboard/procurement/orders/{id}` (or via procurement workspace)

From the detail page:

- View all PO lines with ordered, received, and pending quantities
- See linked GRNs (goods receipt notes)
- Mark PO as sent/acknowledged
- Cancel PO (if no goods received)
- Link to supplier portal view

Goods Receipt Note (GRN)

When materials arrive, a GRN records the actual received quantities against the PO. GRN status values:

Status	Meaning
DRAFT	GRN being filled in
POSTED	GRN confirmed; inventory updated
CANCELLED	GRN voided

On posting a GRN:

1. Stock is credited to the selected warehouse
2. `received_quantity` on each PO line is updated
3. PO status moves to PARTIALLY_RECEIVED or FULLY_RECEIVED
4. A purchase invoice can be generated from the GRN for payment processing

Procurement Workspace Tabs (Full List)

Tab Key	Label	Content
<code>purchase-requests</code>	Purchase Requests	PR list and creation
<code>orders</code>	Purchase Orders	PO list and management
<code>rfq</code>	RFQ	Request for quotation management
<code>deliveries</code>	Deliveries	Goods receipt notes
<code>suppliers</code>	Suppliers	Supplier master data
<code>blanket-agreements</code>	Blanket Agreements	Blanket purchase agreements
<code>reorder-policies</code>	Reorder Policies	Auto-reorder configuration
<code>suggestions</code>	Suggestions	AI-generated procurement suggestions
<code>subcontracting</code>	Subcontracting	Outsourced manufacturing orders
<code>landed-cost</code>	Landed Cost	Import duty, freight, insurance allocation
<code>supplier-portal</code>	Supplier Portal	Self-service portal for suppliers

All tabs require `procurement.view`.

RFQ & Supplier Quotations

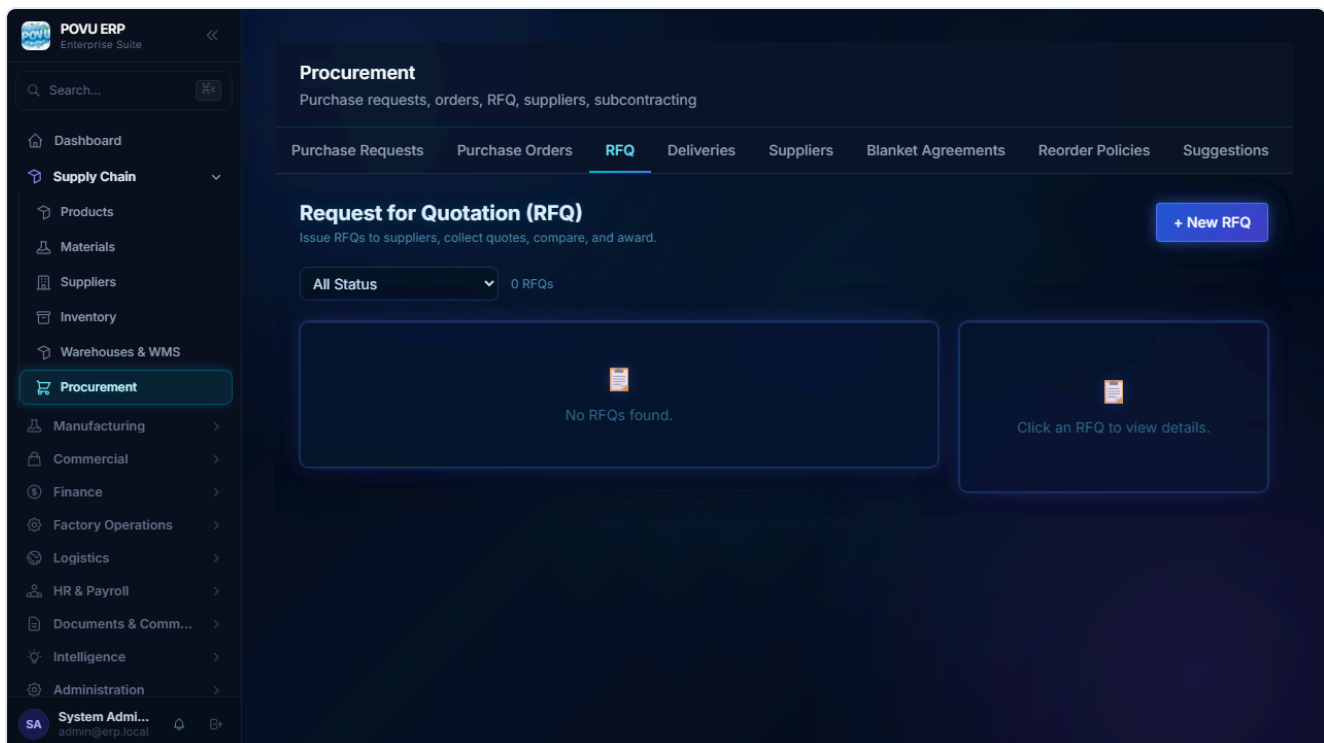
Route: `/dashboard/procurement?tab=rfq`

Tab key: `rfq`

Permission required: `procurement.view`

What It Does

A Request for Quotation (RFQ) is sent to one or more suppliers to obtain pricing before committing to a Purchase Order. Suppliers respond with their quoted prices; the procurement team compares responses and awards the business to the preferred supplier, which then creates a PO.



RFQ tab showing active quotation requests with supplier response status.

The screenshot displays the POVL ERP Enterprise Suite interface. On the left is a navigation menu with options like Dashboard, Supply Chain, Products, Materials, Suppliers, Inventory, Warehouses & WMS, Procurement, Manufacturing, Commercial, Finance, Factory Operations, Logistics, HR & Payroll, Documents & Comm..., Intelligence, Administration, and System Administration. The main area shows the 'Procurement' section with a 'Request for Quotation' card. A 'New RFQ' modal is open, allowing users to create a new Request for Quotation. The modal includes fields for RFQ No., Title, Description, Quantity, Unit, Required By, and Response Deadline, along with 'Create RFQ' and 'Cancel' buttons.

New RFQ modal — title, closing date, material lines, and invited suppliers.

RFQ Workflow

Create RFQ → Send to Suppliers → Receive Responses → Compare → Award → PO Created

RFQ Status Values

Status	Meaning
DRAFT	RFQ being prepared
SENT	RFQ dispatched to suppliers
RESPONSES_RECEIVED	At least one supplier has responded
AWARDED	Business awarded to a supplier; PO being created
CANCELLED	RFQ cancelled
CLOSED	RFQ closed without awarding

RFQ Response Status Values

Status	Meaning
PENDING	Supplier has not yet responded
RECEIVED	Response submitted
ACCEPTED	This response was awarded
REJECTED	Supplier not selected

Creating an RFQ

An RFQ is typically created from a PR or independently. Fields include:

- RFQ number (user-assigned; e.g. RFQ-2026-001)
- Closing date (deadline for supplier responses)
- Materials/items with quantities and target units
- Selected suppliers (one RFQ can go to multiple suppliers)

Each selected supplier receives an individual response slot. Their quoted unit prices, lead times, and notes are recorded against each supplier response.

Comparing Responses

The RFQ comparison view shows all supplier responses side by side:

- Quoted unit price per line
- Total value
- Lead time quoted
- Terms and conditions
- Previous performance rating (if configured)

The procurement officer selects the winning supplier and clicks **Award**. This creates a PO pre-filled with the winning supplier's prices.

Blanket Purchase Agreements

Tab key: blanket-agreements

Blanket Purchase Agreements (BPAs) pre-negotiate pricing with a supplier for a defined period and volume, without committing to specific delivery dates. Call-off orders are raised against the BPA.

BPA status values: DRAFT, ACTIVE, EXPIRED, CANCELLED.

Key fields:

- Agreement number
- Supplier
- Validity period (start date → end date)
- Material/product lines with agreed unit prices
- Cumulative call-off quantity limit

When a PO is raised against a BPA, the agreed price is applied automatically and the cumulative call-off quantity is tracked against the BPA limit.

Reorder Policies

Tab key: `reorder-policies`

Reorder policies define automatic replenishment triggers for materials:

Field	Description
Material	The material this policy governs
Warehouse	The warehouse where the reorder point applies
Reorder Point	Minimum on-hand quantity that triggers a reorder alert
Reorder Quantity	How much to order each time
Lead Time (days)	Days from PO creation to delivery
Preferred Supplier	Supplier pre-selected on auto-generated PRs

When `quantity_on_hand` drops below the Reorder Point, the stock summary shows a `⚠ low` warning and the Procurement Suggestions tab generates a suggested PR.

Procurement Suggestions

Tab key: `suggestions`

The Suggestions tab aggregates:

- Materials below reorder point (from Reorder Policies)
- MRP-generated material requirements (from Planning module)
- AI-generated suggestions based on consumption trends

Each suggestion shows the material, shortfall quantity, suggested reorder quantity, and recommended supplier. Approving a suggestion creates a PR automatically.

Subcontracting

Tab key: subcontracting

Subcontracting orders are used when manufacturing is outsourced to a third-party processor. The subcontract order:

- Specifies the product to be made and the quantity
- Lists the raw materials to send to the subcontractor
- Records the finished goods received back

Subcontracting integrates with production (work orders) and inventory (material issue + goods receipt).

Landed Cost

Tab key: landed-cost

Landed cost allocation apportions import costs (freight, insurance, customs duty, handling) across the material lines of a PO. This ensures the true cost of imported materials is captured in inventory valuation.

Allocation methods:

- **By Value** — proportional to line value
- **By Quantity** — proportional to quantity
- **By Weight** — proportional to material weight
- **Manual** — user specifies amounts per line

After posting landed costs, the material's average cost in inventory is updated to include the allocated import costs.

Deliveries & Goods Receipt

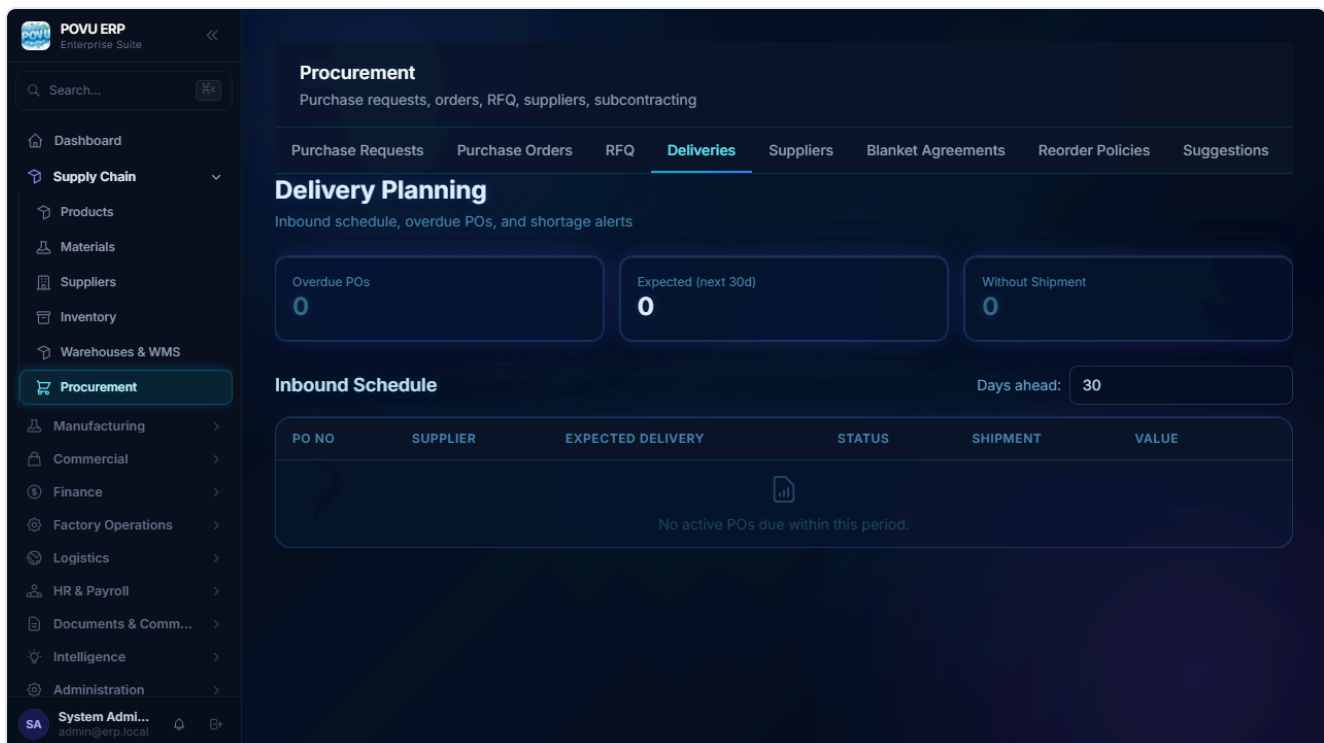
Route: /dashboard/procurement?tab=deliveries

Tab key: deliveries

Permission required: procurement.view

What It Does

The Deliveries tab manages Goods Receipt Notes (GRNs) — the documents that record the physical arrival of materials from suppliers. Posting a GRN credits inventory and updates PO receipt quantities. It also triggers QC inspection requests for incoming materials when configured.



Deliveries tab showing Goods Receipt Notes with status (DRAFT / POSTED / CANCELLED).

GRN Status Values

Status	Meaning
DRAFT	GRN being filled in; inventory not yet updated
POSTED	GRN confirmed; inventory credited; PO updated

Status	Meaning
CANCELLED	GRN voided; no inventory impact

Creating a GRN

A GRN is typically created from the PO detail page by clicking **Receive Delivery**. It can also be created from the Deliveries tab.

GRN Header Fields

Field	Required	Notes
GRN No	Yes	User-assigned; e.g. GRN-2026-001
PO Reference	Yes	Links to the Purchase Order
Supplier	Yes	Auto-populated from PO
Delivery Date	Yes	Actual date goods arrived
Warehouse	Yes	Where goods are being received
Notes	No	Free text; delivery note or driver name

GRN Line Fields

Each GRN line corresponds to a PO line:

Field	Required	Notes
PO Line	Yes	Links to specific PO line
Material / Product	—	Auto-populated from PO line
Ordered Qty	—	Read-only from PO line
Received Qty	Yes	Actual quantity received; can be less than ordered (partial delivery)
Lot Number	No	Supplier lot for traceability
Expiry Date	No	Expiry date on the received lot
Unit Cost	No	Override if different from PO price

Partial Deliveries

Suppliers may deliver in multiple shipments. Each delivery is a separate GRN against the same PO:

- `received_quantity` on the PO line accumulates across all GRNs
- PO status shows `PARTIALLY_RECEIVED` until all lines are fully received
- The `pending_quantity` on each PO line = `ordered_quantity - received_quantity`

Posting a GRN

Posting the GRN:

1. Validates that received quantities are within allowed tolerance
2. Creates an **inventory ENTRY movement** — credits `quantity_on_hand` in the warehouse
3. Assigns the lot number and expiry date to the stock record
4. Updates PO line `received_quantity`
5. Updates PO status to `PARTIALLY_RECEIVED` or `FULLY_RECEIVED`
6. Creates a **QC inspection request** (if incoming QC is configured for this material)
7. Optionally generates a **purchase invoice** for accounts payable

Once posted, a GRN cannot be edited. To correct a posting, a stock adjustment must be raised.

Import Shipment Integration

For international imports, the GRN links to the Logistics module's shipment and container records.

When goods arrive from the import shipment:

1. The logistics team records the arrival in **Logistics → Arrivals**
2. Customs clearance status is updated
3. The warehouse team then creates GRNs to receive the goods into inventory
4. Landed costs are allocated to the GRN lines

Import Shipment Status values:

Status	Meaning
<code>PLANNED</code>	Shipment being organised
<code>BOOKED</code>	Vessel / cargo space confirmed
<code>CARGO_LOADED</code>	Container loaded at origin port
<code>IN_TRANSIT</code>	On the vessel at sea
<code>ARRIVED_PORT</code>	Arrived at Mombasa port

Status	Meaning
CUSTOMS_HOLD	Held by KRA for inspection
CUSTOMS_CLEARED	KRA clearance obtained
OUT_FOR_DELIVERY	In transit from port to warehouse
DELIVERED	Received at warehouse
CANCELLED	Shipment cancelled

Suppliers

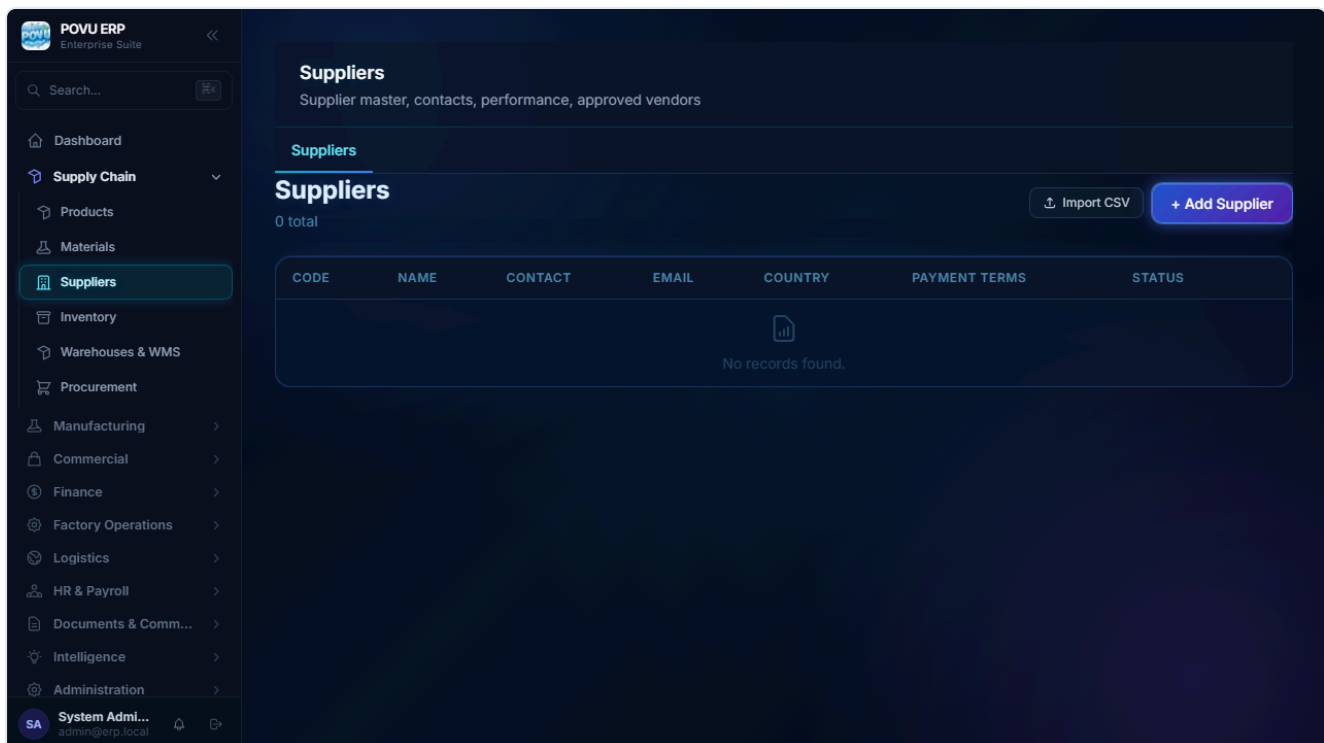
Route: `/dashboard/procurement?tab=suppliers`

Tab key: `suppliers`

Permission required: `procurement.view`

What It Does

The Suppliers tab manages the supplier master data — the registry of all vendors from whom materials and products are purchased. Supplier records are referenced across procurement (PRs, POs, RFQs), quality (incoming inspections, certificates), and logistics (import shipments).



Supplier master list showing code, name, country, contact, and active/inactive status.

Supplier List

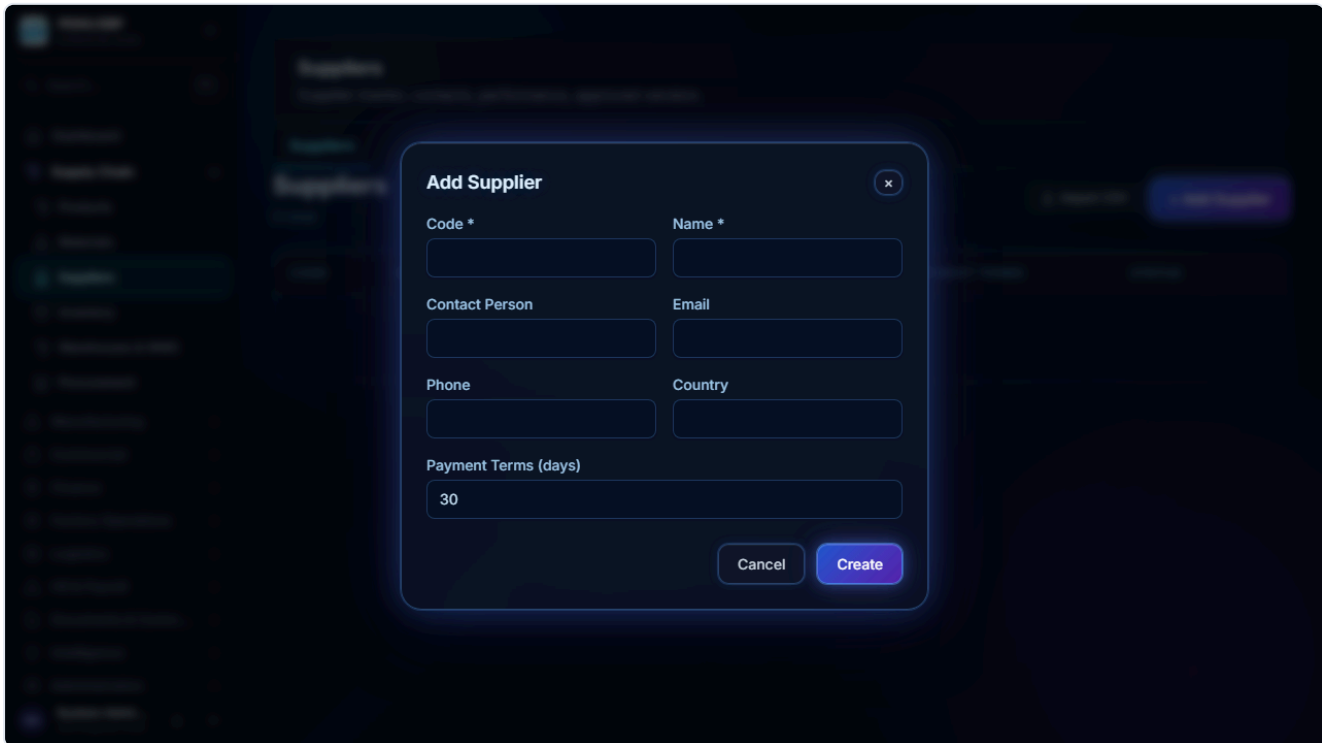
Shows all suppliers with:

- Supplier name
- Supplier code (monospace)
- Country

- Contact information
- Active/Inactive status
- Link to supplier detail

Add Supplier Form

Click **+ Add Supplier** to open the create dialog.



Add Supplier form: Code, Name, Contact Person, Email, Phone, Country, and Payment Terms.

Fields shown in the form:

Field	Required	Default	Notes
Code	Yes	—	Internal supplier code (e.g. SUP-001)
Name	Yes	—	Legal business name
Contact Person	No	—	Primary contact name
Email	No	—	Contact email address
Phone	No	—	Contact phone number
Country	No	—	Country of registration
Payment Terms (days)	No	30	Default net payment terms in days

Supplier Record Fields

All fields stored per supplier record (full schema including CSV-importable fields):

Field	Required	Description
<code>code</code>	Yes	Internal supplier code (e.g. <code>SUP-001</code>)
<code>name</code>	Yes	Legal business name
<code>contact_person</code>	No	Primary contact person name
<code>email</code>	No	Email address
<code>phone</code>	No	Phone number
<code>address</code>	No	Full street address
<code>city</code>	No	City
<code>country</code>	No	Country of registration
<code>tax_id</code>	No	VAT / tax registration number
<code>payment_terms_days</code>	No	Default payment terms in days (default: 30)
<code>is_active</code>	No	Whether supplier is available for new orders (default: true)
<code>preferred_payment_method</code>	No	<code>bank</code> , <code>cash</code> , or <code>mpesa</code>
<code>mpesa_phone_number</code>	No	M-Pesa number (required when method is <code>mpesa</code>)
<code>supplier_category</code>	No	Internal category / segment label
<code>qualification_status</code>	No	AVL status — see Supplier Approval Status below
<code>risk_level</code>	No	Procurement risk rating — <code>LOW</code> , <code>MEDIUM</code> , <code>HIGH</code> , <code>CRITICAL</code>
<code>approved_from</code>	No	Approval start date (YYYY-MM-DD)
<code>approved_until</code>	No	Approval expiry date (YYYY-MM-DD)

Importing Suppliers via CSV

Click **Import** on the Suppliers page to bulk-load supplier records from a CSV file.

Steps:

1. Download the CSV template from the Import dialog.

2. Fill in one supplier per row.
3. Upload the file and click **Import**.
4. Review the validation summary — rows with errors are skipped; valid rows are created.

CSV column reference:

Column	Required	Accepted values / format
code	Yes	Unique code, e.g. SUP-001
name	Yes	Legal business name
contact_person	No	Contact person name
email	No	Valid email address
phone	No	Phone number
address	No	Full street address
city	No	City name
country	No	Country name
tax_id	No	VAT / tax ID number
payment_terms_days	No	Integer, e.g. 30 (default: 30)
is_active	No	true or false (default: true)
preferred_payment_method	No	bank, cash, or mpesa
mpesa_phone_number	No	M-Pesa phone number (required if method is mpesa)
supplier_category	No	Free-text category label
qualification_status	No	PENDING, APPROVED, CONDITIONAL, SUSPENDED, REJECTED
risk_level	No	LOW, MEDIUM, HIGH, CRITICAL
approved_from	No	Date in YYYY-MM-DD format
approved_until	No	Date in YYYY-MM-DD format

Columns marked **Yes** are mandatory — rows missing `code` or `name` will be rejected.

Supplier Approval Status

The `qualification_status` field controls whether a supplier can be used on orders:

Status	Meaning
PENDING	Awaiting qualification review
APPROVED	Fully qualified; on the Approved Vendor List (AVL)
CONDITIONAL	Approved with restrictions or pending re-qualification
SUSPENDED	Temporarily blocked from new orders
REJECTED	Permanently disqualified

Only **APPROVED** suppliers can be selected on Purchase Orders. **CONDITIONAL** suppliers may require override approval depending on configuration.

Approved Vendor List (AVL)

Each material can have one or more approved suppliers. The AVL enforces that materials are only purchased from qualified vendors. When raising a PO line, the supplier field filters to show only AVL-approved suppliers for that material.

Supplier Portal

Tab key: `supplier-portal`

Route: `/dashboard/supplier-portal/page` (embedded)

The Supplier Portal gives external suppliers (or procurement staff acting on their behalf) a view of:

- Open POs awaiting acknowledgment
- Delivery schedules
- Outstanding GRNs
- Invoice status

Access is controlled per supplier and requires supplier-specific credentials or a guest token.

Supplier Performance

From the supplier detail page, procurement managers can review:

- On-time delivery rate
- Quality pass rate (from QC incoming inspections)
- Price variance vs. quoted prices
- Outstanding invoice amounts

These metrics feed into the RFQ comparison and supplier selection decisions.

Blanket Agreements, Reorder Policies & Suggestions

This chapter is a reference supplement to the RFQ chapter. See [03-rfq.md](#) for full documentation of Blanket Agreements, Reorder Policies, and Procurement Suggestions.

Quick reference:

Feature	Tab Key	Purpose
Blanket Agreements	<code>blanket-agreements</code>	Pre-negotiated prices for a period; call-off orders
Reorder Policies	<code>reorder-policies</code>	Automatic reorder point triggers per material + warehouse
Suggestions	<code>suggestions</code>	AI + MRP generated procurement recommendations
Subcontracting	<code>subcontracting</code>	Outsourced manufacturing orders
Landed Cost	<code>landed-cost</code>	Allocate import costs across PO lines

Blanket Agreements Tab

Route: `/dashboard/procurement?tab=blanket-agreements`

Blanket Agreements tab showing pre-negotiated agreements with supplier, validity period, and status.

POVL ERP
Enterprise Suite

Search...

Dashboard

Supply Chain

Products

Materials

Suppliers

Inventory

Warehouses & WMS

Procurement

Manufacturing

Commercial

Finance

Factory Operations

Logistics

HR & Payroll

Documents & Comm...

Intelligence

Administration

System Admin...
admin@erp.local

Procurement

Purchase request

Blanket Purchase Agreement

Long-term price agreement

Active Agreements: 0

All Status

New Blanket Agreement

BPA No. *
BPA-2026-001

Supplier ID *
UUID

Description
e.g. Palm Oil FY2026

Agreed Unit Price *
0.00

Currency
KES

Agreed Quantity (blank = open)

Unit
KG

Valid From *
mm/dd/yyyy

Valid To *
mm/dd/yyyy

Payment Terms
Net 30

Notes

Create

Cancel

Agreements

Reorder Policies

Suggestions

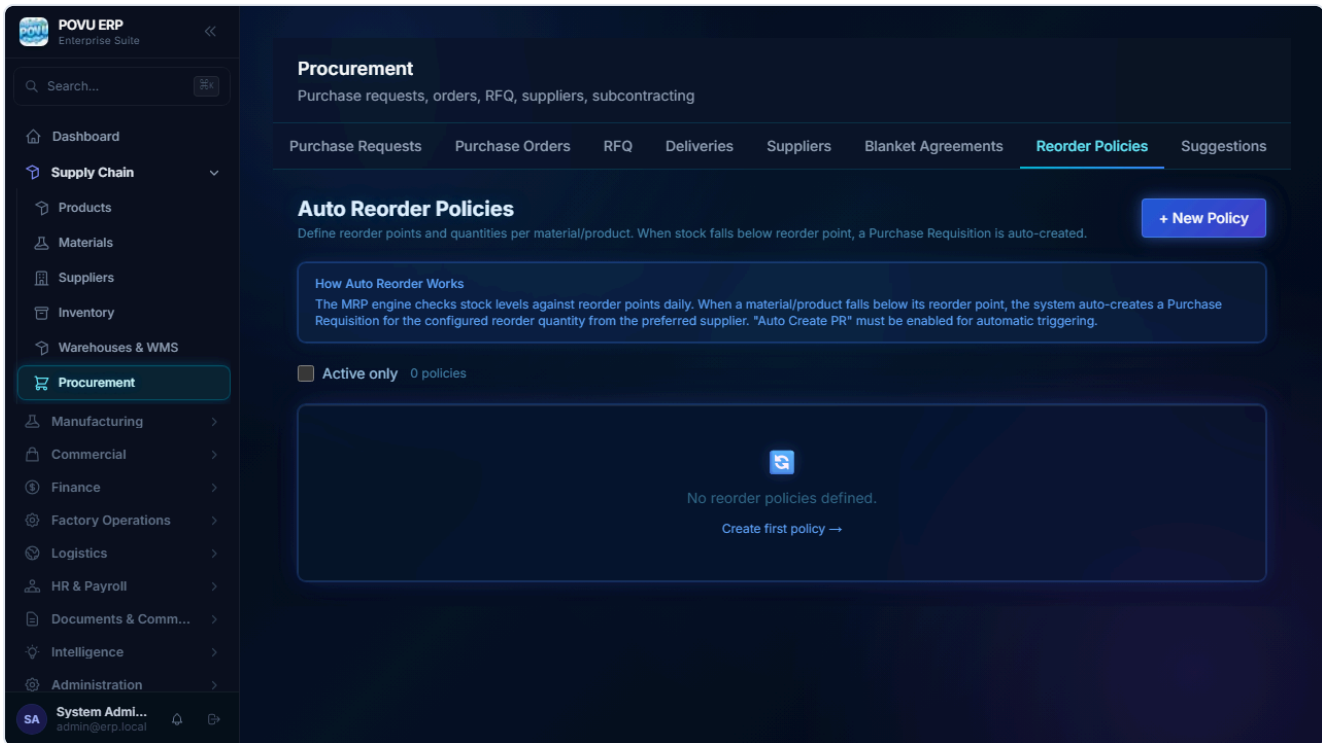
+ New Agreement

Total Contracted Value
KES 0

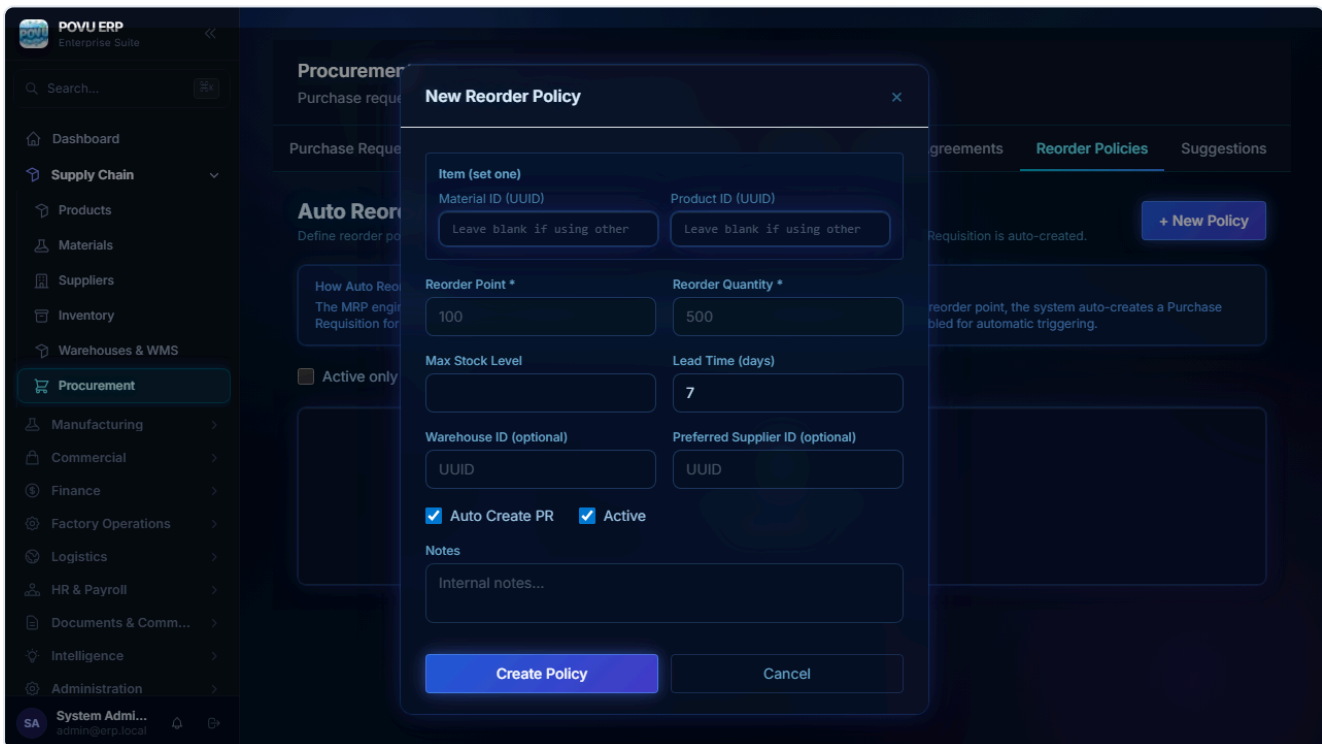
New Blanket Agreement modal — Supplier, Agreement No, Validity dates, and committed quantity/value.

Reorder Policies Tab

Route: `/dashboard/procurement?tab=reorder-policies`



Reorder Policies tab listing automatic replenishment rules per material and warehouse.



New Reorder Policy modal — Material, Warehouse, Reorder Point, Reorder Quantity, Lead Time, Preferred Supplier.

Inventory Stock Ledger

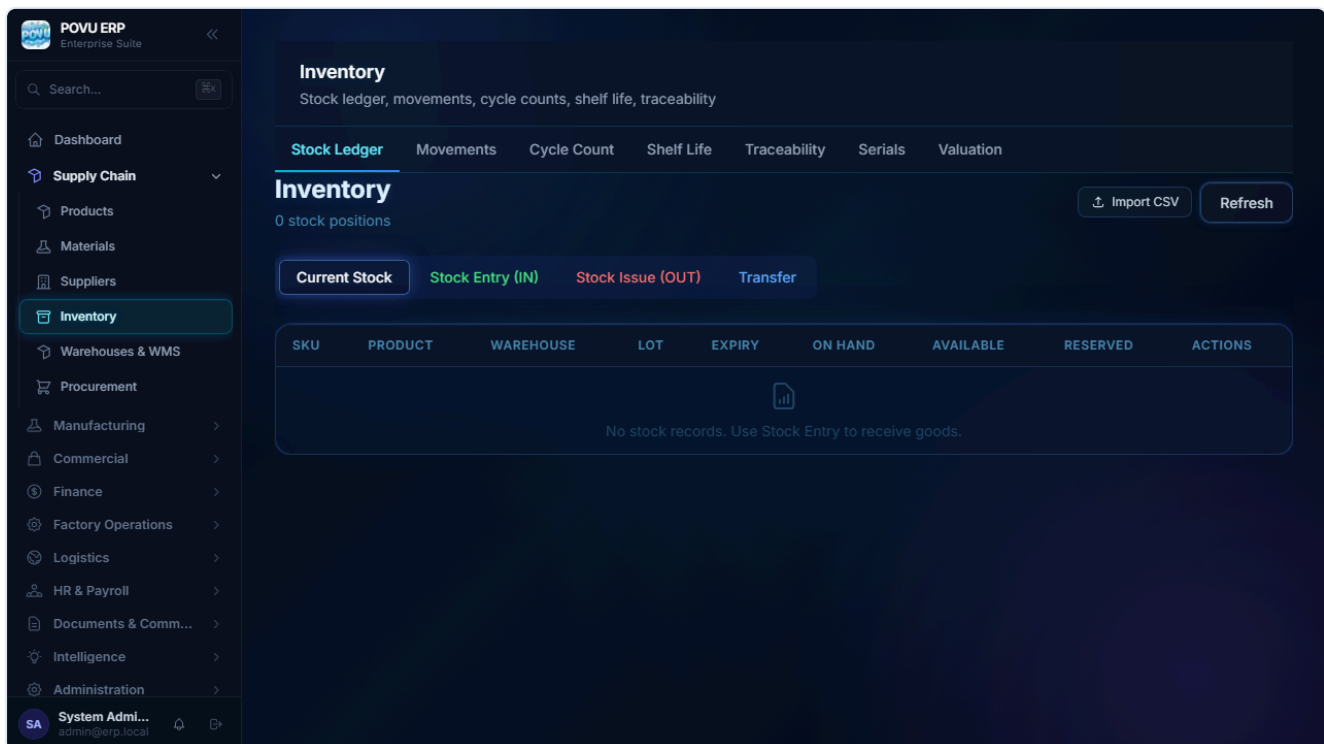
Route: `/dashboard/inventory` (default tab: Stock Ledger)

Permission required: `inventory.view`

Workspace tabs: Stock Ledger, Movements, Cycle Count, Shelf Life, Traceability, Serials, Valuation

What It Does

The Inventory module manages finished goods stock positions across all warehouses. Each stock record (position) represents a unique combination of product + warehouse + lot number. The inventory page provides four operational sub-tabs: stock summary, stock entry (receive), stock issue (dispatch), and inter-warehouse transfer.



Stock Ledger showing on-hand, available, and reserved quantities with reorder alerts.

Stock Summary (Current Stock Tab)

Tab: `stock` — `data-testid="inventory-stock-table"`

Shows all stock positions with real-time quantities.

Table Columns

Column	Field	Notes
SKU	<code>product_sku</code>	Monospace
Product	<code>product_name</code>	Product description
Warehouse	<code>warehouse_code</code> – <code>warehouse_name</code>	Composite display
Lot	<code>lot_number</code>	"no lot" shown in grey if null
Expiry	<code>expiry_date</code>	Blank if no expiry
On Hand	<code>quantity_on_hand</code>	Bold red + ⚠ low if <code>is_below_reorder = true</code>
Available	<code>quantity_available</code>	Emerald text; = On Hand minus Reserved
Reserved	<code>quantity_reserved</code>	Quantity committed to open sales orders
Actions	—	Adjust / Delete (permission-gated)

Quantity Relationships

```
quantity_on_hand = quantity_available + quantity_reserved
```

- `quantity_available` — what can be picked for new orders
- `quantity_reserved` — held for open but not yet dispatched sales orders
- `is_below_reorder` — true when `quantity_on_hand < reorder_point` (from Reorder Policies)

Reorder Alert

Stock positions below their reorder point show a bold red quantity and a ⚠ low suffix. This signals procurement to raise a PR.

Stock Entry (IN) — Receiving Stock

Tab: `entry` (green label)

Used to receive finished goods into a warehouse. Typically triggered after a production batch is completed or a direct purchase receipt outside the procurement module.

Stock Entry form for receiving goods: product, warehouse, quantity, lot number, expiry date, and reference.

Stock Entry Fields

Field	Label	Required	Backend field	Notes
<code>product_id</code>	Product	Yes	<code>product_id</code>	Select from products master (SKU — Name)
<code>warehouse_id</code>	Warehouse	Yes	<code>warehouse_id</code>	Select from warehouses (Code — Name)
<code>quantity</code>	Quantity	Yes	<code>quantity</code>	Decimal; step 0.001; must be > 0
<code>unit_cost</code>	Unit Cost (Rp)	No	<code>unit_cost</code>	Cost per unit for inventory valuation
<code>lot_number</code>	Lot Number	No	<code>lot_number</code>	e.g. <code>LOT-2026-001</code>
<code>expiry_date</code>	Expiry Date	No	<code>expiry_date</code>	Date picker
<code>reference</code>	Reference #	Yes	<code>reference</code>	Default pre-filled as <code>GR-YYYY-MM-DD</code> ; identifies the source document
<code>notes</code>	Notes	No	<code>notes</code>	Free text

Permission check: Button disabled if user lacks `inventory.receive_all` and is out of scope for the selected warehouse. Error message: "You can view this warehouse but cannot receive stock in this

scope."

On success: `quantity_on_hand` increases; a RECEIPT movement is created in the ledger; returns to Stock tab.

Stock Issue (OUT) — Dispatching Stock

Tab: `issue` (red label)

Used to issue finished goods from a warehouse. Typically used for sales order fulfilment or production material issue outside the main work order flow.

Stock Issue form for dispatching goods: product, warehouse, quantity, lot number, and reference.

Stock Issue Fields

Field	Label	Required	Backend field	Notes
<code>product_id</code>	Product	Yes	<code>product_id</code>	Select from products master
<code>warehouse_id</code>	Warehouse	Yes	<code>warehouse_id</code>	Select from warehouses
<code>quantity</code>	Quantity	Yes	<code>quantity</code>	Decimal; step 0.001; must be > 0
<code>lot_number</code>	Lot Number (optional)	No	<code>lot_number</code>	Specify lot for FEFO picking

Field	Label	Required	Backend field	Notes
reference	Reference #	Yes	reference	Default SO-YYYY-MM-DD
notes	Notes	No	notes	Free text

Insufficient stock guard: The API returns an error if `quantity > quantity_available`. The UI shows a toast: "Insufficient Stock" with the API's message. An amber warning box in the form states: "The system will reject this if available stock is insufficient."

Permission check: Button disabled if user lacks `inventory.dispatch_all` and is out of scope for the selected warehouse.

On success: `quantity_on_hand` decreases; an ISSUE movement is created; returns to Stock tab.

Transfer — Moving Stock Between Warehouses

Tab: `transfer` (blue label)

Transfers stock between two warehouses of the same company.

The screenshot displays the 'Inventory' section of the FMCG ERP Supply Chain module. The left sidebar shows the navigation menu with 'Inventory' selected. The main content area is titled 'Inventory' and includes tabs for 'Stock Ledger', 'Movements', 'Cycle Count', 'Shelf Life', 'Traceability', 'Serials', and 'Valuation'. The 'Stock Ledger' tab is active, showing '0 stock positions'. Below this, there are buttons for 'Current Stock', 'Stock Entry (IN)', 'Stock Issue (OUT)', and 'Transfer'. The 'Transfer' button is highlighted. The 'Transfer Stock Between Warehouses' form contains the following fields:

- Product ***: A dropdown menu with the placeholder text '— Select product —'.
- From Warehouse ***: A dropdown menu with the placeholder text '— Select warehouse —'.
- To Warehouse ***: A dropdown menu with the placeholder text '— Select warehouse —'.
- Quantity ***: A text input field.
- Lot Number (optional)**: A text input field.
- Reference # ***: A text input field containing the value 'TRF-2026-05-19'.
- Notes**: A text area for additional information.

Transfer form: select product, source warehouse, destination warehouse, quantity, and reference.

Transfer Fields

Field	Label	Required	Backend field	Notes
<code>product_id</code>	Product	Yes	<code>product_id</code>	Select from products master
<code>from_warehouse_id</code>	From Warehouse	Yes	<code>from_warehouse_id</code>	Source warehouse
<code>to_warehouse_id</code>	To Warehouse	Yes	<code>to_warehouse_id</code>	Destination warehouse
<code>quantity</code>	Quantity	Yes	<code>quantity</code>	Decimal; step 0.001; must be > 0
<code>lot_number</code>	Lot Number (optional)	No	<code>lot_number</code>	Specify lot to maintain FEFO tracking
<code>reference</code>	Reference #	Yes	<code>reference</code>	Default TRF-YYYY-MM-DD
<code>notes</code>	Notes	No	<code>notes</code>	Free text

Permission check: Requires `inventory.transfer_all` OR `canPerformInScope` on both the source and destination warehouses. Button disabled with tooltip if insufficient permissions.

On success: source warehouse `quantity_on_hand` decreases; destination increases; two TRANSFER movements created (OUT from source, IN to destination); returns to Stock tab.

Adjusting Stock

From the Stock Summary tab, users with `inventory.adjust` (in scope) or `inventory.adjust_all` see an **Adjust** button per row.

Adjustment Modal

Field	Required	Notes
New Quantity	Yes	The corrected on-hand quantity (not the delta)
Reason	Yes	e.g. "Physical count correction", "Damaged goods write-off"

The adjustment creates an ADJUSTMENT movement. The delta (new – old) is shown in green (positive) or red (negative) before submission.

Deleting a Stock Record

From the Stock Summary tab, users with `inventory.delete` (in scope) or `inventory.delete_all` see a **Delete** button.

Constraint: A stock record can only be deleted when `quantity_on_hand = 0`. If the quantity is non-zero, the delete modal shows a warning: "You must adjust the quantity to zero before deleting this record."

Blocked error (HTTP 409): If the stock record is referenced by other data (e.g. open sales order reservations), deletion is blocked with a structured error listing the blocking references.

CSV Import

The `ImportModal` (module: `inventory_stock`) allows bulk stock positions to be imported via CSV. Use this for initial data load or period-end corrections.

CSV columns (inspect `ImportModal` configuration for exact headers — the module key is `inventory_stock`).

Inventory Workspace Tabs

Tab Key	Label	Content
<code>stock</code>	Stock Ledger	Stock summary, entry, issue, transfer
<code>movements</code>	Movements	Full movement ledger
<code>cycle-count</code>	Cycle Count	Physical count sheets and reconciliation
<code>shelf-life</code>	Shelf Life	Expiry management and FEFO alerts
<code>traceability</code>	Traceability	Forward and backward lot trace
<code>serials</code>	Serials	Serial number tracking
<code>valuation</code>	Valuation	Inventory value reports

All tabs require `inventory.view`.

Demo Data — Inventory Seed (I1–I7)

The system ships with FMCG inventory seed data populated across all inventory tabs. This data represents a realistic starting state for demonstration, training, and testing.

Inventory
Stock ledger, movements, cycle counts, shelf life, traceability

Stock Ledger Movements Cycle Count Shelf Life Traceability Serials Valuation

Inventory
5 stock positions

Import CSV Refresh

Current Stock Stock Entry (IN) Stock Issue (OUT) Transfer

SKU	PRODUCT	WAREHOUSE	LOT	EXPIRY	ON HAND	AVAILABLE	RESERVED	ACTIONS
POVU-LD-1L	POVU Liquid Detergent 1L	FG-WH — Finished Goods Warehouse	no lot	—	4,500	4,500	0	Adjust Delete
POVU-FS-500ML	POVU Fabric Softener 500ml	FG-WH — Finished Goods Warehouse	no lot	—	2,800	2,800	0	Adjust Delete
POVU-DW-500ML	POVU Dishwashing Liquid 500ml	FG-WH — Finished Goods Warehouse	no lot	—	3,600	3,600	0	Adjust Delete
POVU-SC-750ML	POVU Surface Cleaner 750ml	FG-WH — Finished Goods Warehouse	no lot	—	2,300	2,300	0	Adjust Delete
POVU-HS-200ML	POVU Hand Sanitizer 200ml	FG-WH — Finished Goods Warehouse	no lot	—	7,500	7,500	0	Adjust Delete

Inventory workspace showing seeded stock positions across PROD-WH and FG-WH warehouses.

Inventory
Stock ledger, movements, cycle counts, shelf life, traceability

Stock Ledger Movements Cycle Count Shelf Life Traceability Serials Valuation

Inventory
5 stock positions

Import CSV Refresh

Current Stock Stock Entry (IN) Stock Issue (OUT) Transfer

SKU	PRODUCT	WAREHOUSE	LOT	EXPIRY	ON HAND	AVAILABLE	RESERVED	ACTIONS
POVU-LD-1L	POVU Liquid Detergent 1L	FG-WH — Finished Goods Warehouse	no lot	—	4,500	4,500	0	Adjust Delete
POVU-FS-500ML	POVU Fabric Softener 500ml	FG-WH — Finished Goods Warehouse	no lot	—	2,800	2,800	0	Adjust Delete
POVU-DW-500ML	POVU Dishwashing Liquid 500ml	FG-WH — Finished Goods Warehouse	no lot	—	3,600	3,600	0	Adjust Delete
POVU-SC-750ML	POVU Surface Cleaner 750ml	FG-WH — Finished Goods Warehouse	no lot	—	2,300	2,300	0	Adjust Delete
POVU-HS-200ML	POVU Hand Sanitizer 200ml	FG-WH — Finished Goods Warehouse	no lot	—	7,500	7,500	0	Adjust Delete

Stock Ledger tab with seeded stock levels, lot numbers, and reorder alerts.

Seed Warehouses

Code	Name	Purpose
PROD-WH	Production Warehouse	Raw materials and in-process stock
FG-WH	Finished Goods Warehouse	Completed product stock

Seeded Products (5) and Materials (7)

Products and materials are sourced from the production master data seed (TASK-015). Each inventory record links to these items via SKU/material code.

I1 — Lots and Stock Positions

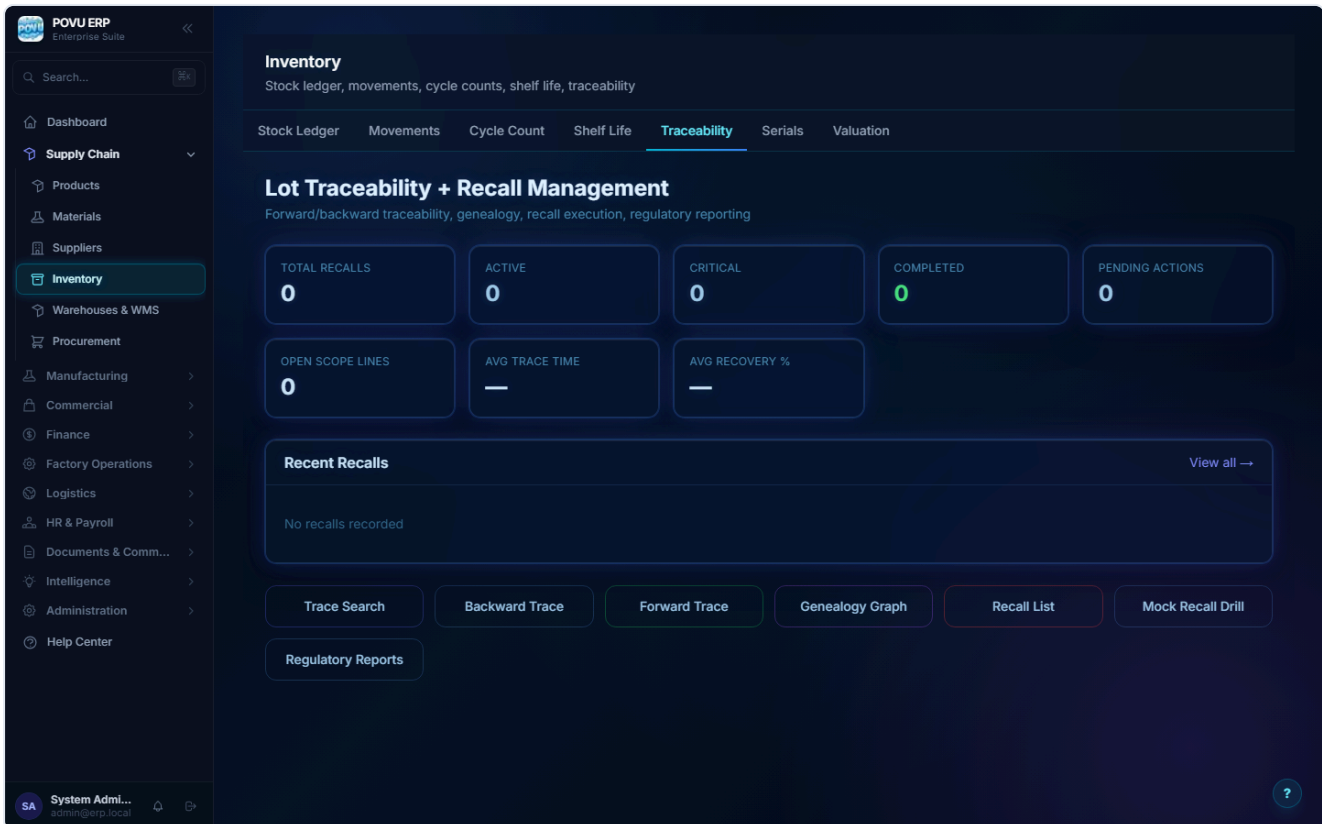
Opening lot records and stock balances seeded for all 5 products and 7 materials across both warehouses. Each lot has a `lot_number`, `expiry_date`, and opening quantity.

I2 — WMS Zones and Storage Locations

WMS zones (PROD-WH and FG-WH) and bin locations seeded. See the WMS manual for zone/location details.

I3 — Trace Events for Lot Genealogy

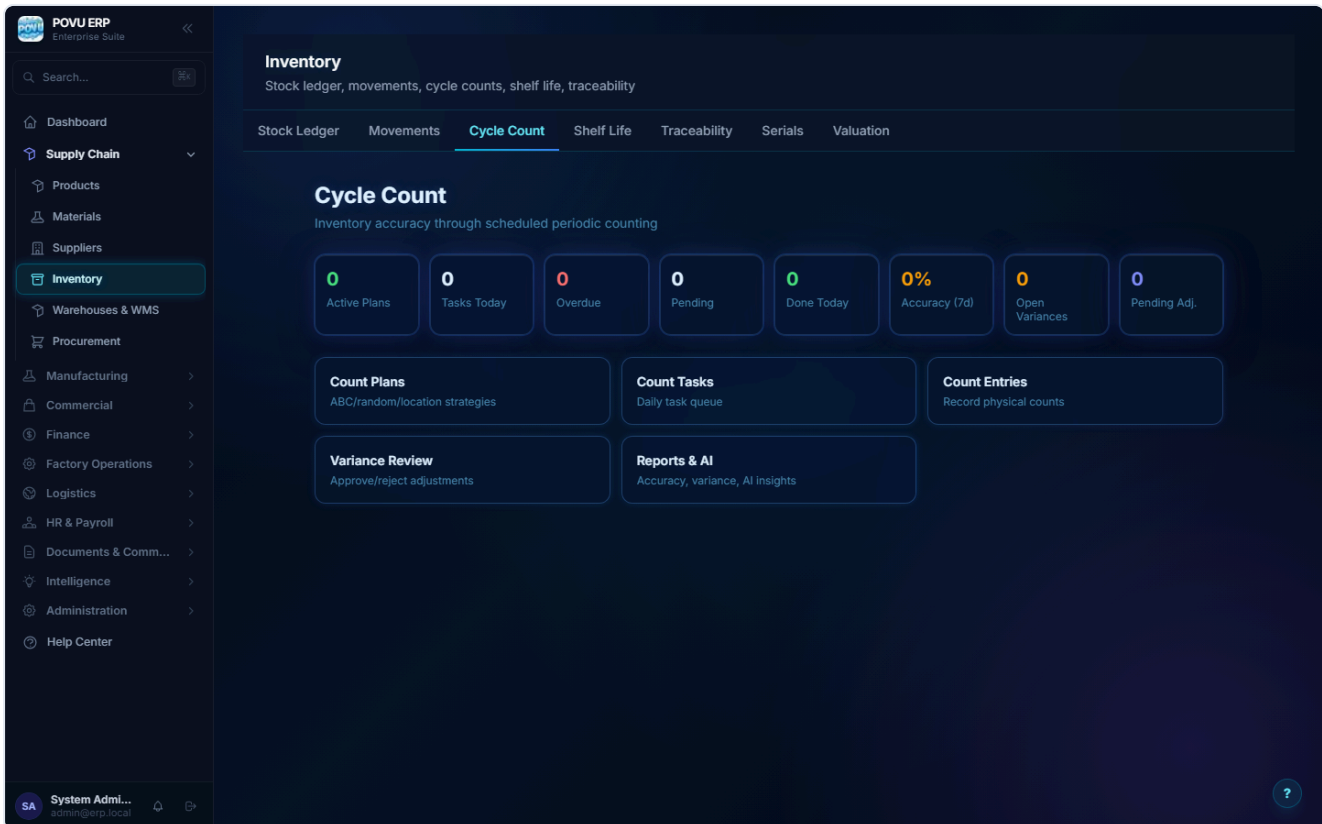
Trace events seeded to support forward and backward lot traceability. Each lot has receipt, issue, and production consumption events recorded.



Traceability tab showing lot genealogy chain from raw material receipt through production consumption.

14 — Cycle Count Plans

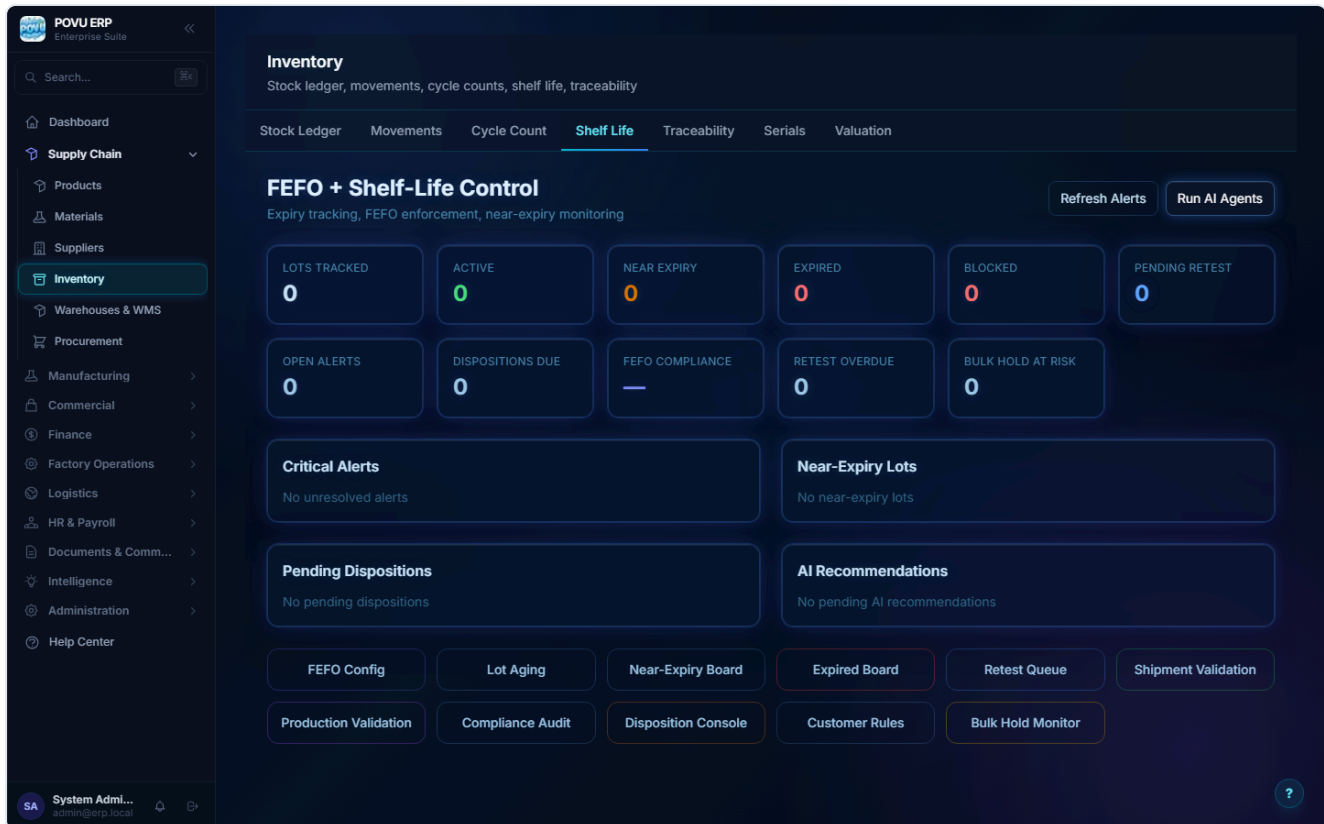
Cycle count plans seeded for both warehouses with planned count dates and item assignments.



Cycle Count tab showing seeded count plans with status and item coverage.

15 — Shelf Life Profiles and Alerts

Lot shelf life profiles seeded with `expiry_date` , `best_before_date` , and `days_remaining` calculations. Shelf life alerts generated for lots approaching expiry thresholds.



Shelf Life tab showing seeded lots with expiry alerts and FEFO ranking.

I6 — Demand Forecasts

Demand forecasts seeded using `ForecastModelType.PROPHET` (implemented as local statsmodels Holt-Winters). See the AI & Automation manual for details on the forecasting engine.

Forecast records include:

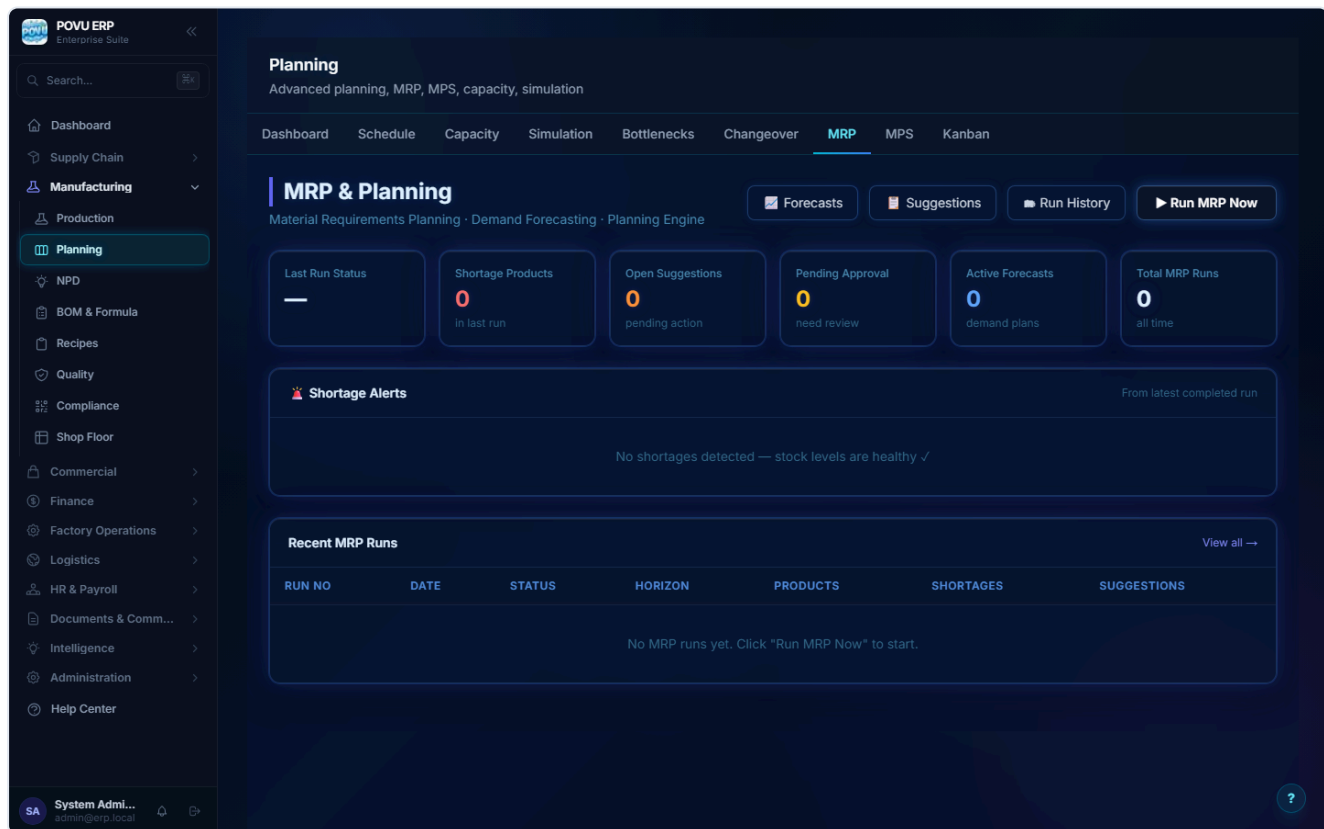
- Forecast model type: `PROPHET` (maps to local Holt-Winters — no external API required)
- Forecast status: `COMPLETED`
- Period type: `MONTHLY`
- Forecast lines per product with predicted demand quantities

I7 — MRP Run, Results, Suggestions, and Exceptions

An MRP run (`MRP-SEED-001`) is seeded with a 90-day planning horizon:

Data type	Count
MRP run	1 (<code>MRP-SEED-001</code> , status <code>COMPLETED</code>)
MRP results	5 (one per product)
MRP suggestions	Multiple per product

Data type	Count
MRP exceptions	Includes shortage flag for <code>POVU-HS</code>



MRP page showing seeded MRP run with demand forecasts, results, and shortage alerts.

Shortage flag: Product `POVU-HS` has `shortage_flag = true` in the MRP result, indicating projected stock-out within the planning horizon.

Shelf Life Management

Tab: `shelf-life`

The Shelf Life tab manages lot expiry, FEFO picking order, and near-expiry alerts.

Shelf Life Alert Types

Alert type	Trigger
Near expiry	Lot within the configured warning threshold days of expiry
Expired	Lot past expiry date
Best before breach	Lot past best-before date (quality advisory; not a hard block)

FEFO Picking

Lots are ranked by expiry date. The oldest-expiry lot is issued first (First Expired, First Out). This ranking is enforced by `shelf_life_service.py` which reads all lots — FEFO must see the complete lot list to rank correctly.

Traceability

Tab: `traceability`

Forward and backward lot tracing across the supply chain:

Direction	Coverage
Backward (upstream)	From finished lot → production batch → raw material receipt
Forward (downstream)	From raw material lot → which work orders consumed it → which finished batches

Trace events record: receipt, issue, production consumption, inter-warehouse transfer, and adjustment.

Cycle Count

Tab: `cycle-count`

Cycle count plans schedule periodic physical counts of stock locations. Each plan defines:

- Count date
- Warehouses in scope
- Items or item categories to count
- Assigned counter

After counting, actual quantities are entered and variance reports are generated. Discrepancies trigger stock adjustments.

Valuation

Tab: `valuation`

Inventory valuation uses FIFO cost layers. Each receipt creates a cost layer. Issues are costed using the oldest available cost layer first.

Metric	Description
Total inventory value (KES)	Sum of all on-hand stock at FIFO cost
Value by warehouse	Per-warehouse inventory value
Value by product	Per-product inventory value
Cost layer details	Unit cost and quantity per open FIFO layer

Inventory Movements & Traceability

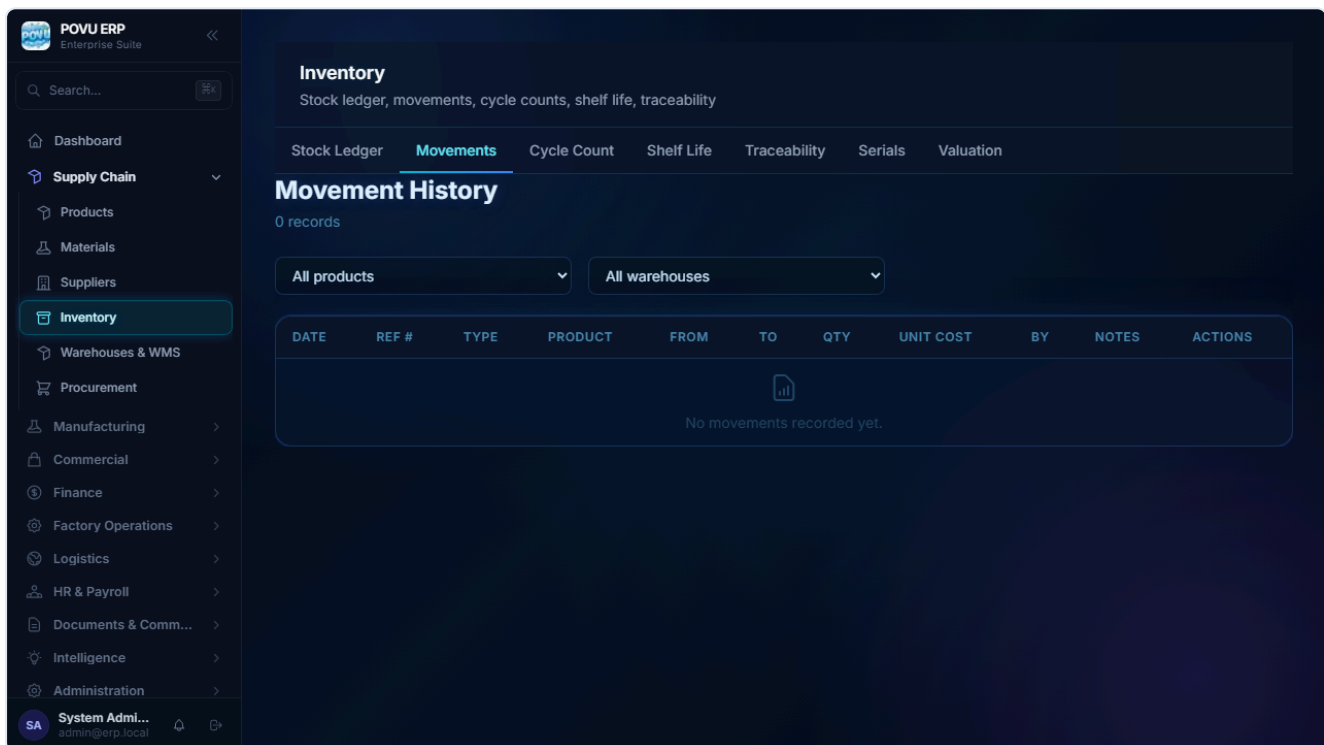
Route: /dashboard/inventory (various tabs)

Permission required: inventory.view

Movements Tab

Tab key: movements

Route: /dashboard/movements/page



Movements ledger showing all stock changes with movement type, quantity, reference, and timestamp.

Every change to inventory — receipt, issue, transfer, adjustment, quarantine — creates an immutable movement record in the ledger. Movements are the audit trail of all stock changes.

Movement Types

Type	Direction	Created By
RECEIPT	IN	Stock Entry form, GRN posting
ISSUE	OUT	Stock Issue form, sales order dispatch

Type	Direction	Created By
TRANSFER_IN	IN	Transfer (destination side)
TRANSFER_OUT	OUT	Transfer (source side)
ADJUSTMENT	IN or OUT	Stock Adjustment modal
PRODUCTION_ISSUE	OUT	Work order material issuance
PRODUCTION_RECEIPT	IN	Work order finished goods receipt
QUARANTINE	OUT (blocked)	WMS quarantine action
QUARANTINE_RELEASE	IN (unblocked)	WMS quarantine release

Movement Record Fields

Field	Description
movement_no	System-generated reference
movement_type	Type from list above
product_sku + product_name	Product affected
warehouse_name	Warehouse affected
quantity	Signed quantity (positive = IN, negative = OUT)
lot_number	Lot affected (if applicable)
reference	Source document reference (PO no, SO no, etc.)
notes	Free text notes
created_at	Timestamp of the movement
created_by	User who created it

Filtering Movements

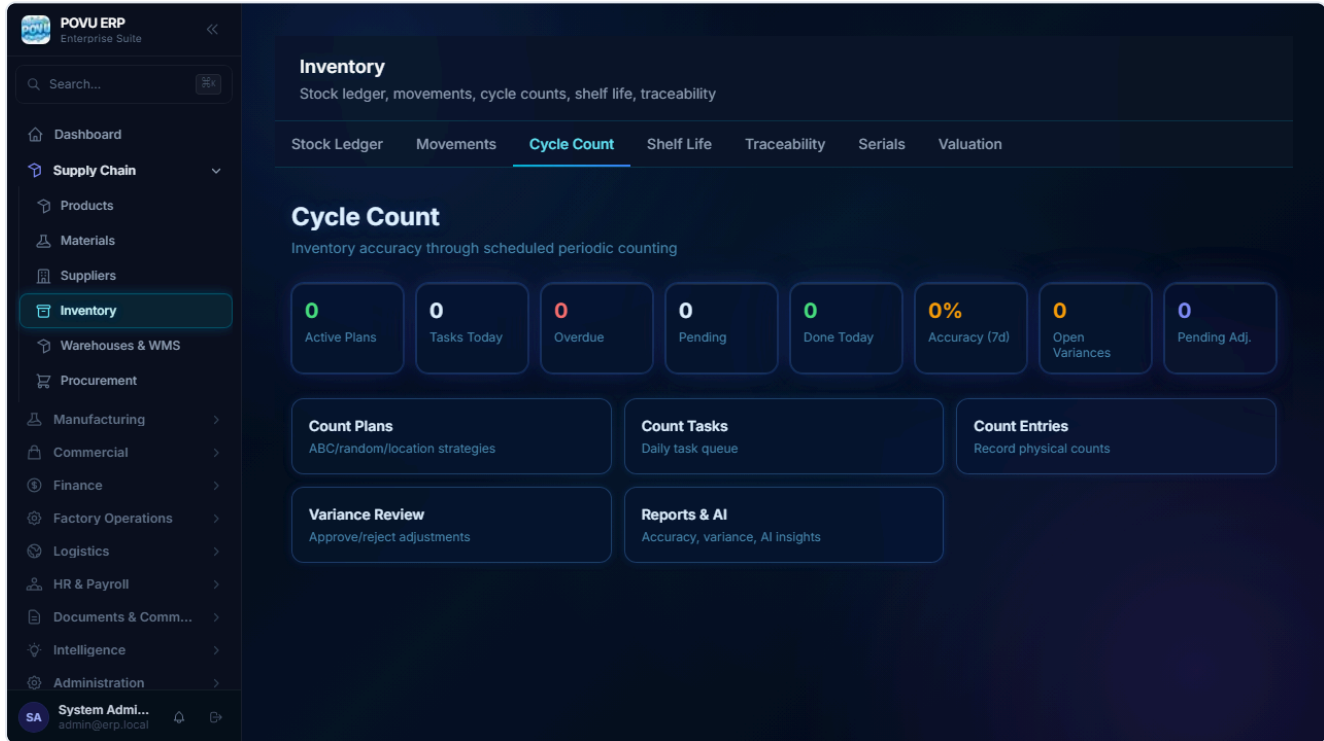
The Movements tab supports filtering by:

- Date range
- Movement type
- Product
- Warehouse

Cycle Count Tab

Tab key: cycle-count

Route: /dashboard/cycle-count/page



Cycle Count tab showing count sheets with status (DRAFT, IN_PROGRESS, POSTED, CANCELLED).

Cycle counting is the process of periodically counting a subset of inventory items to verify the accuracy of the stock ledger without doing a full warehouse shutdown.

Cycle Count Process

1. **Create Count Sheet** — select products and/or locations to count
2. **Print / Export** — generate count sheets for warehouse staff
3. **Enter Counts** — staff record physical counts on the sheet
4. **Reconcile** — system compares physical counts to book quantities
5. **Post Adjustments** — approve and post variances as ADJUSTMENT movements

Count Status Values

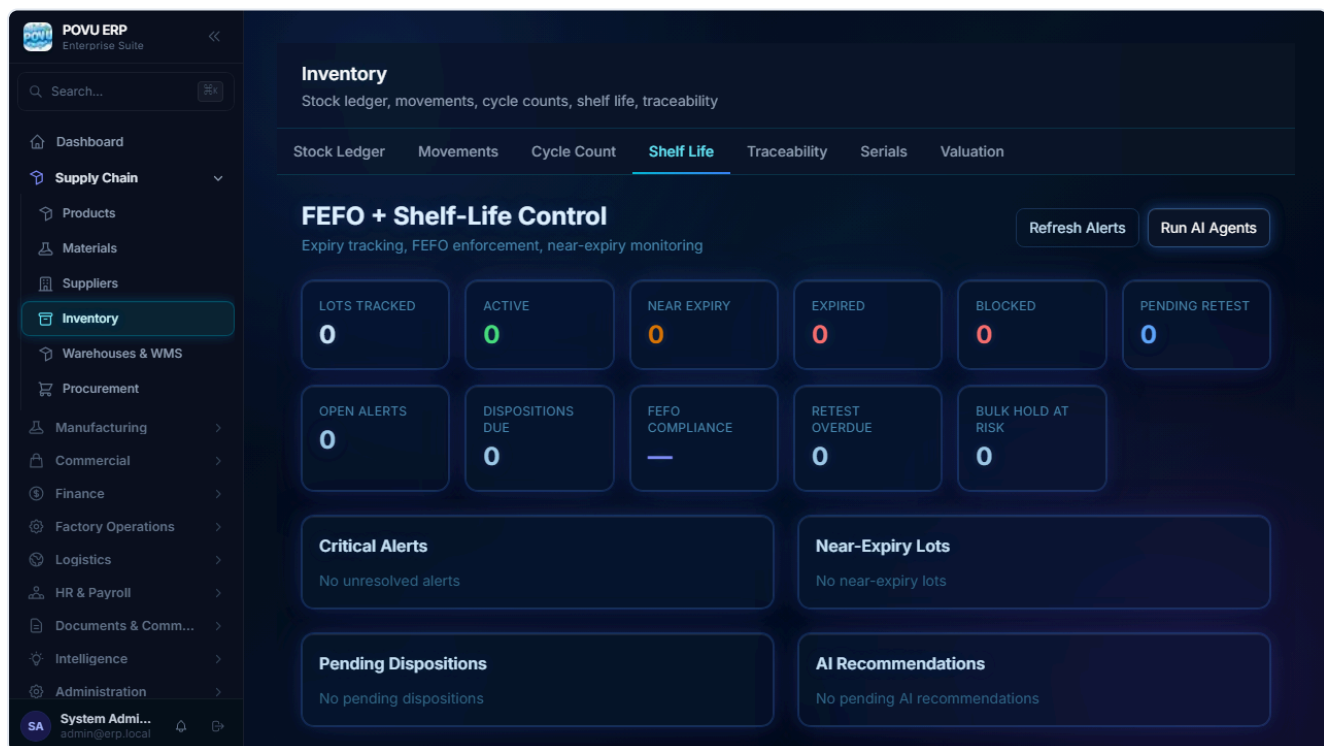
Status	Meaning
DRAFT	Count sheet created
IN_PROGRESS	Counting underway

Status	Meaning
POSTED	Adjustments applied; ledger updated
CANCELLED	Count voided

Shelf Life Tab

Tab key: shelf-life

Route: /dashboard/shelf-life/page



Shelf Life tab showing expiring stock positions grouped by urgency with FEFO ranking.

Manages expiry dates across all stock positions.

Key Views

View	Description
Expiring within 30 days	Stock positions approaching expiry
Expiring within 90 days	Medium-term expiry planning
Already expired	Expired stock requiring write-off
FEFO ranking	First-Expiry-First-Out picking priority order

FEFO Enforcement

When issuing stock, the system prioritises lot numbers with the earliest expiry date. If `lot_number` is not specified on the issue, the system automatically selects the FEFO lot.

Traceability Tab

Tab key: `traceability`

Route: `/dashboard/traceability/page`

Provides full forward and backward traceability for any product or lot number. See the [Batch & Lots chapter](#) for the traceability model.

Forward Trace

From a production batch number or supplier lot: which customers received it?

Backward Trace

From a sales order or customer: which supplier lots contributed to the product?

Serials Tab

Tab key: `serials`

Route: `/dashboard/inventory/serials/page`

Serial number tracking for high-value products that require individual unit identification (e.g. equipment, appliances). Each serial number:

- Is linked to a specific product and lot
 - Has a status: IN_STOCK / SOLD / RETURNED / SCRAPPED
 - Has a full movement history
-

Valuation Tab

Tab key: `valuation`

Route: `/dashboard/inventory/valuation/page`

Inventory valuation reports the financial value of stock on hand. Valuation methods:

Method	Description
FIFO	First In, First Out — uses the cost of the oldest stock
Weighted Average Cost (WAC)	Average of all receipts for the product
Standard Cost	Uses a pre-defined standard cost per unit

The valuation report shows:

- Stock value per product
- Stock value per warehouse
- Total inventory value at the company level
- Comparison to prior period

This feeds directly into the Finance module's balance sheet for inventory asset reporting.

Warehouses

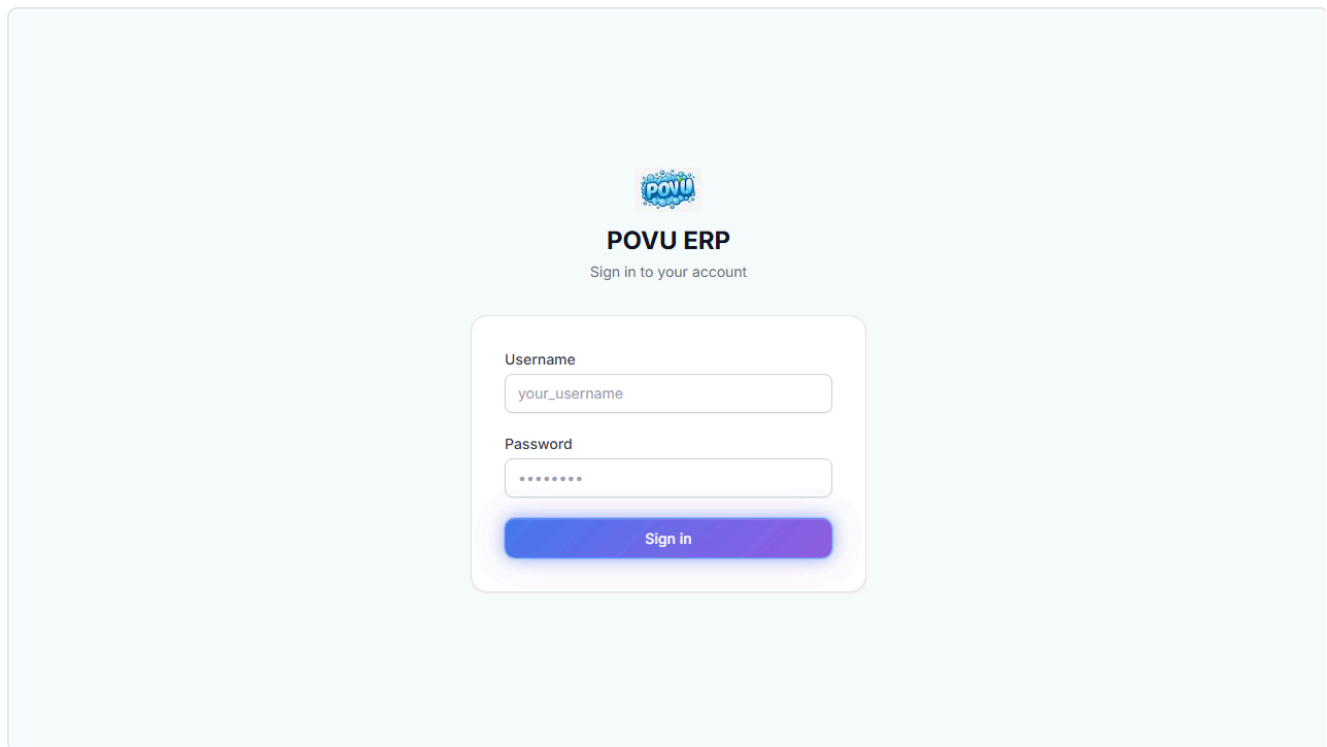
Route: `/dashboard/warehouses`

Permission required: `inventory.view`

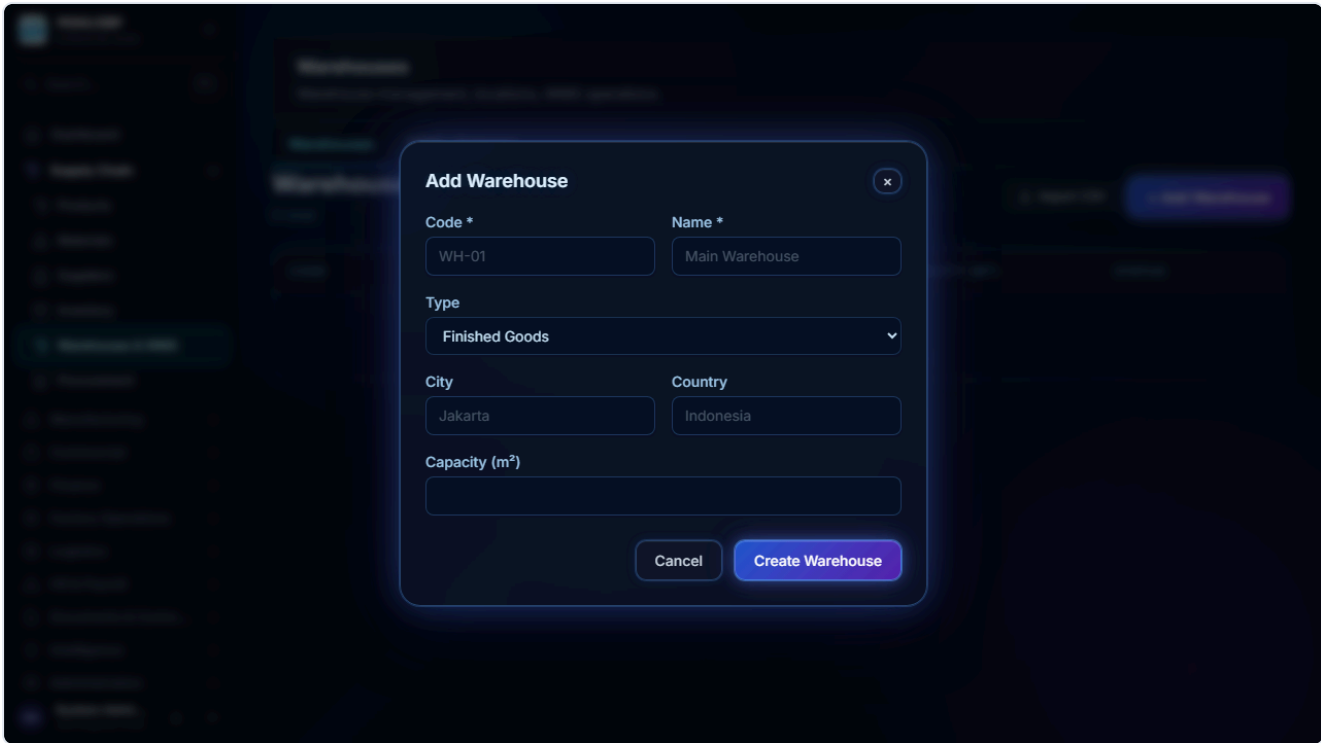
Workspace tabs: Warehouses, WMS

What It Does

The Warehouses module manages the physical warehouse master data — the locations where stock is held. Warehouse records are referenced by inventory positions, GRNs, transfers, WMS zones, and logistics arrivals.



Warehouse list showing code, name, type badge, city, country, capacity, and status.



Add Warehouse modal — Code, Name, Type, City, Country, Capacity (m²) fields.

Warehouse List

Shows all warehouses with:

- Warehouse code (monospace)
- Name
- Type (badge)
- City
- Country
- Capacity (m²)
- Status (Active / Inactive)
- Delete button (permission-gated)

Warehouse Record Fields

Create / Edit Form

Field	Label	Required	Notes
<code>code</code>	Warehouse Code	Yes	Short identifier; e.g. <code>WH-NBI-01</code>

Field	Label	Required	Notes
<code>name</code>	Warehouse Name	Yes	Full descriptive name
<code>warehouse_type</code>	Type	Yes	See type values below
<code>is_active</code>	Active	No	Default <code>true</code>
<code>city</code>	City	No	City where warehouse is located
<code>country</code>	Country	No	Country
<code>capacity_sqm</code>	Capacity (m²)	No	Floor area in square metres

Warehouse Type Values

Type	Description
<code>FINISHED_GOODS</code>	Stores finished products ready for dispatch
<code>RAW_MATERIAL</code>	Stores input materials for production
<code>TRANSIT</code>	Intermediate holding for goods in transit between locations
<code>RETURNS</code>	Holds returned goods pending inspection or disposition

Warehouse Scoping

Warehouse records are scoped to the tenant company. Users with `inventory.view` see only the warehouses within their permitted scope.

Some inventory actions (receive, issue, transfer, adjust) require the user to have permissions scoped to the specific warehouse:

- `inventory.receive_all` — receive into any warehouse
- `inventory.dispatch_all` — issue from any warehouse
- `inventory.transfer_all` — transfer between any warehouses
- `inventory.adjust_all` — adjust stock in any warehouse
- `canPerformInScope("warehouse", warehouse_id, action)` — scope-limited equivalent

WMS Tab

Tab key: `wms`

The Warehouses page embeds the WMS module as a second tab. See [10-wms.md](#) for full WMS documentation.

Deleting a Warehouse

Delete is only available to users with the appropriate permission. A warehouse with associated stock positions, WMS zones, or open documents cannot be deleted — a constraint error is returned.

Warehouse Management System (WMS)

Route: `/dashboard/wms`

Permission required: `inventory.view`

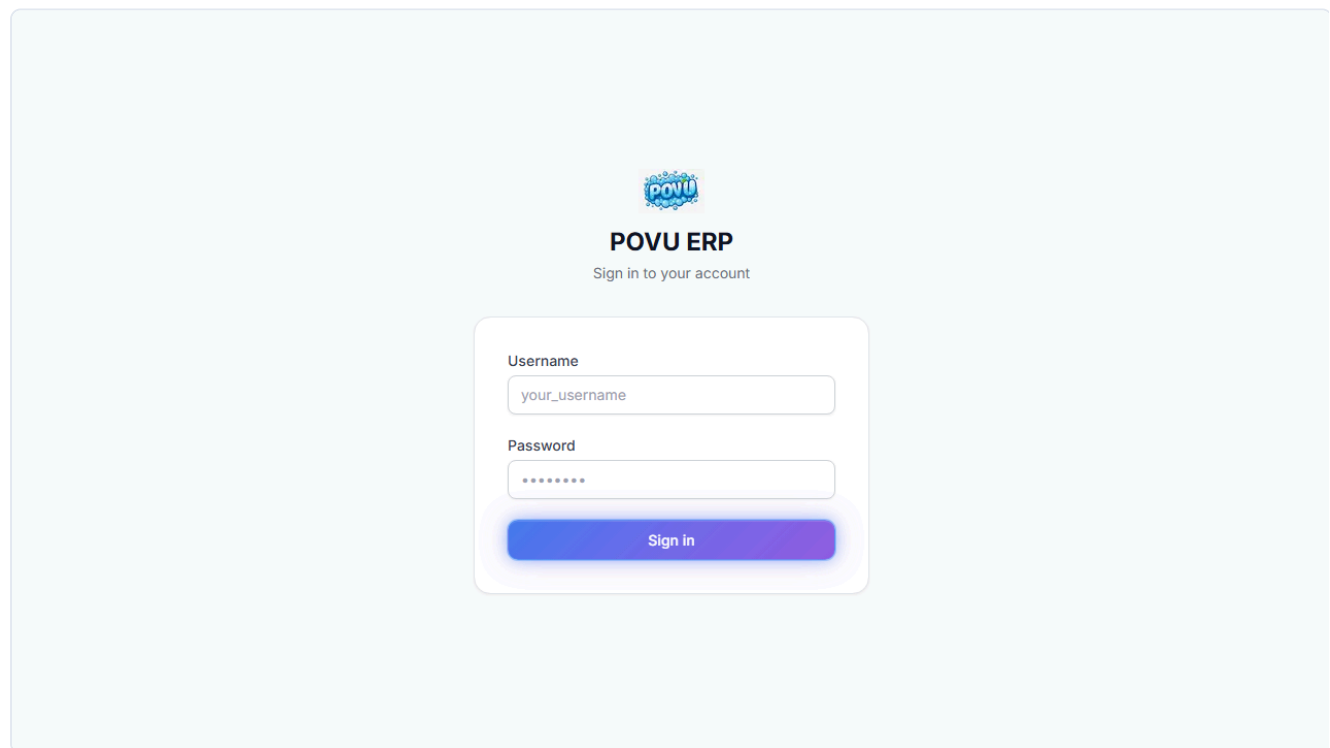
Workspace tabs: Zones, Locations, Handling Units, Pick Waves, Quarantine

What It Does

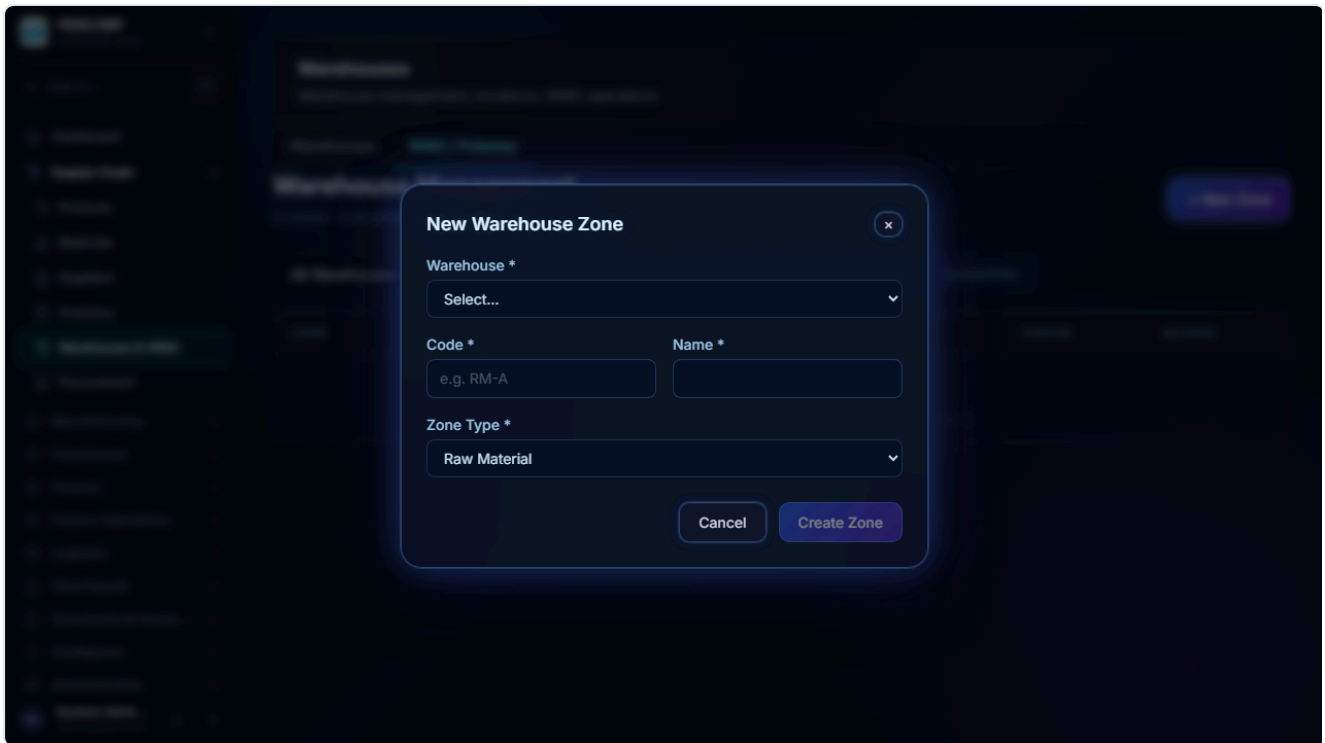
WMS provides sub-warehouse organisation: zones within a warehouse, bin locations within zones, handling unit tracking, wave picking, and quarantine management. It sits on top of the inventory ledger — WMS locations track where within a warehouse stock physically resides.

Zones Tab

Tab key: `zones`



Zones tab showing warehouse zones with type badges (RAW_MATERIAL, FINISHED_GOODS, QUARANTINE, etc.).



New Warehouse Zone modal — Warehouse, Zone Code, Zone Name, and Zone Type fields.

Zones are areas within a warehouse with a specific function (raw material, staging, quarantine, etc.).

Zone Type Values

Type	Description
RAW_MATERIAL	Input material storage
SEMI_FINISHED	WIP / intermediate goods
FINISHED_GOODS	Completed product storage
QUARANTINE	Blocked / suspect stock
RETURNS	Customer returns holding
STAGING	Pick, pack, and dispatch staging

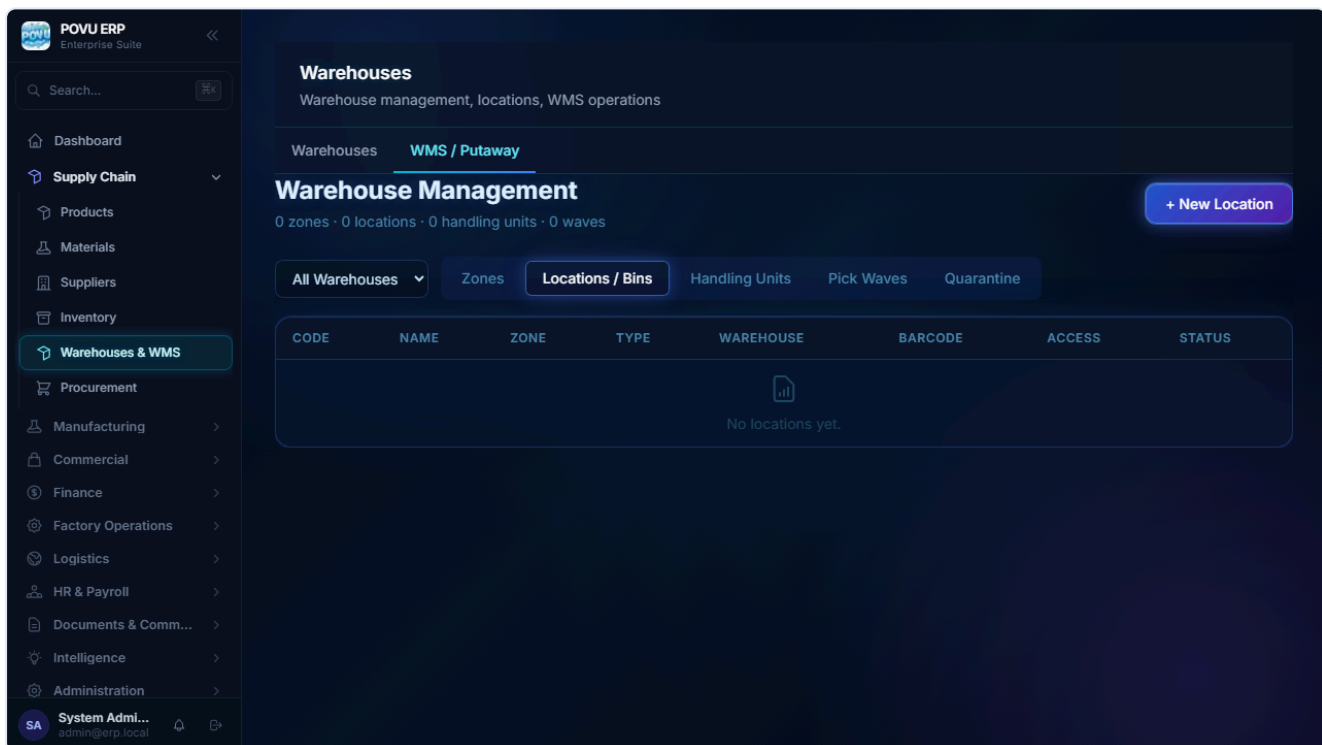
Zone Create Form

Field	Label	Required	Notes
warehouse_id	Warehouse	Yes	Select from warehouses list
code	Zone Code	Yes	Short identifier; e.g. FG-A

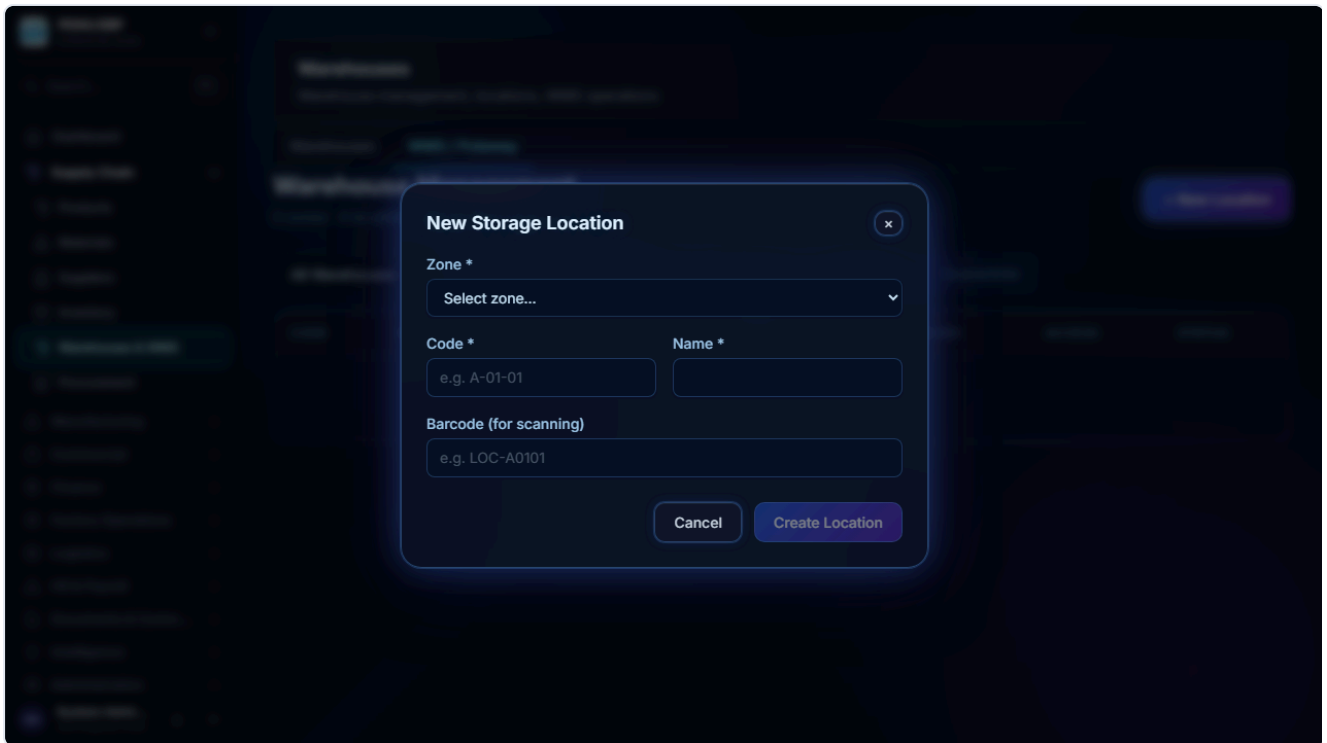
Field	Label	Required	Notes
name	Zone Name	Yes	Descriptive name
zone_type	Zone Type	Yes	See type values above
is_active	Active	No	Default <code>true</code>

Locations Tab

Tab key: `locations`



Locations tab showing bin locations with zone, barcode, blocked status, and active status.



New Storage Location modal — Zone, Location Code, Location Name, and Barcode fields.

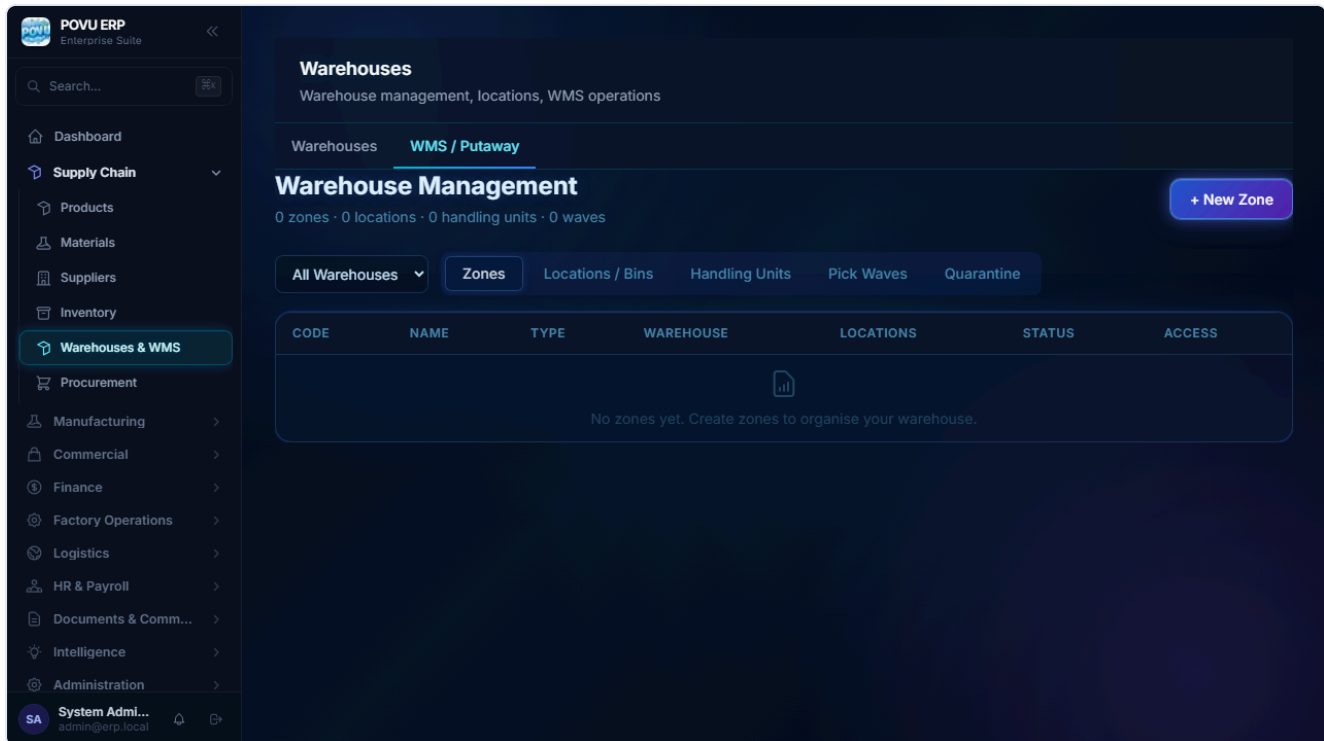
Bin locations are the leaf-level storage positions within a zone (rack, shelf, bin address).

Location Create Form

Field	Label	Required	Notes
<code>zone_id</code>	Zone	Yes	Select from zones list
<code>code</code>	Location Code	Yes	e.g. A-01-03 (aisle-rack-level)
<code>name</code>	Location Name	Yes	Descriptive label
<code>barcode</code>	Barcode	No	Scannable barcode for the location
<code>is_active</code>	Active	No	Default <code>true</code>
<code>is_blocked</code>	Blocked	No	Prevents picks/puts when blocked

Handling Units Tab

Tab key: `handlingUnits`



Handling Units tab listing pallets, totes, and cartons with license plate, location, and status.

Handling units (HUs) group individual stock items under a single licence plate for tracking pallets, totes, or cartons.

Handling Unit List Columns

Column	Description
License Plate	Unique HU identifier (scannable)
Type	HU type (PALLET, TOTE, CARTON, etc.)
Warehouse	Warehouse where HU is located
Location	Bin location within the warehouse
Items	Count of stock lines in the HU
Status	Current HU status
Access	Link to HU detail

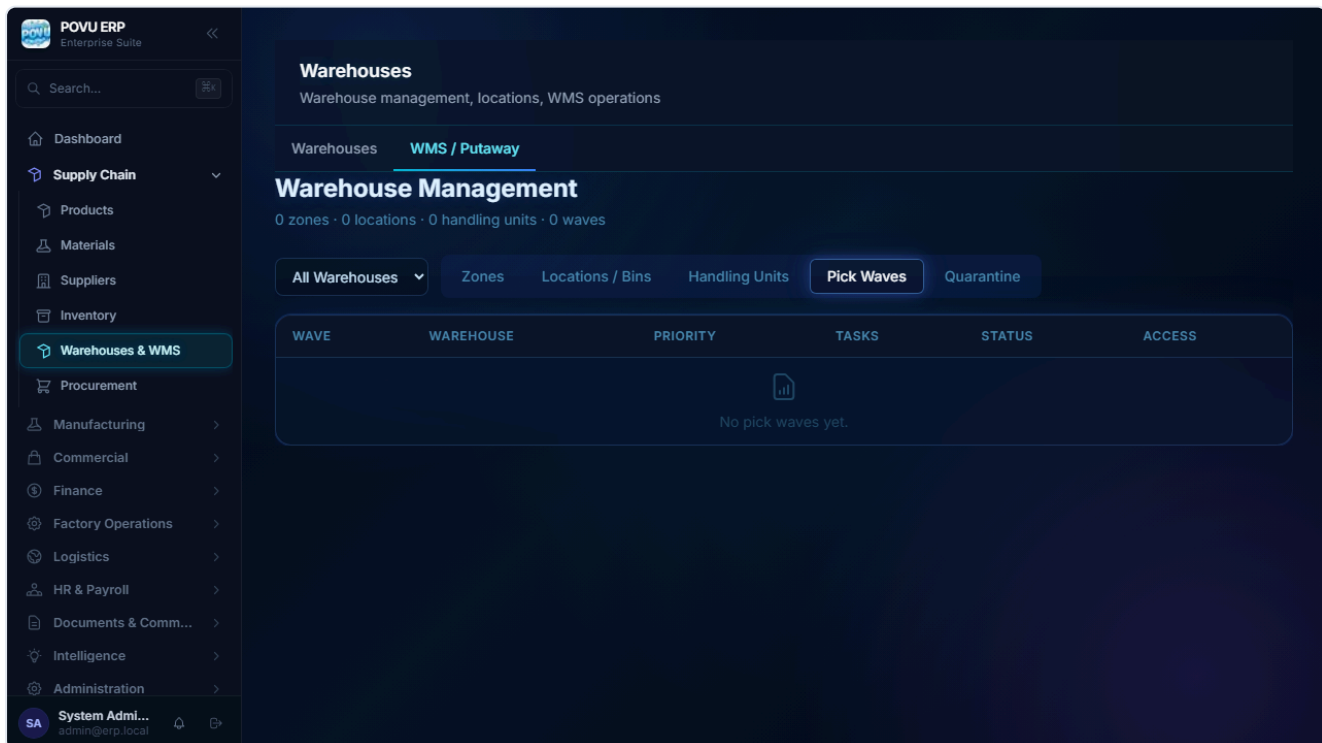
HU Status Badge Colours

Status	Colour
OPEN , DRAFT , PENDING , IN_PROGRESS , RELEASED	Blue

Status	Colour
CLOSED , COMPLETED , PICKED , PACKED , SHIPPED	Green
ON_HOLD , CANCELLED , VOID , CONSUMED	Red

Pick Waves Tab

Tab key: `pickWaves`



Pick Waves tab showing batched picking waves with task count, priority, and status.

Wave picking batches multiple pick tasks across multiple orders into a single wave for efficient picking routes.

Pick Wave List Columns

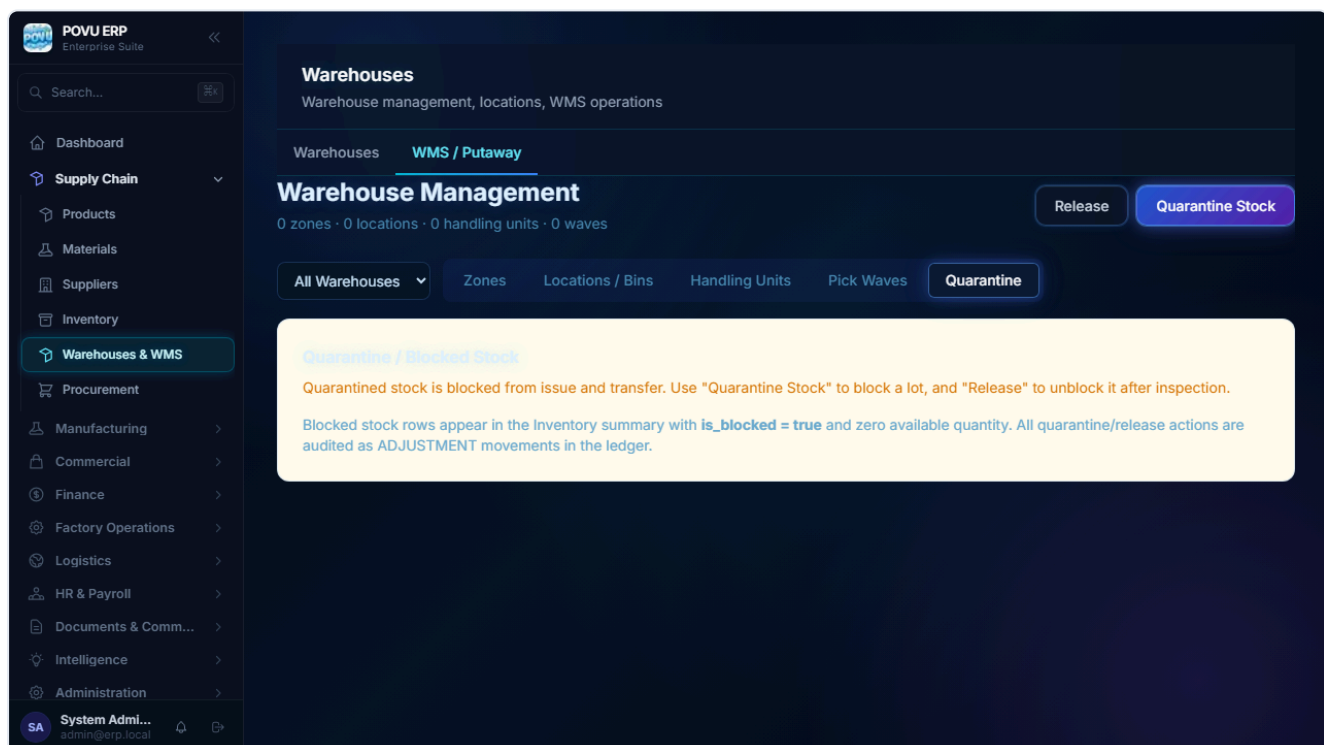
Column	Description
Wave	<code>wave_no</code> — wave reference number
Warehouse	Warehouse the wave belongs to
Priority	Pick priority (higher = more urgent)
Tasks	<code>task_count</code> — number of individual pick tasks

Column	Description
Status	Current wave status
Access	Link to wave detail

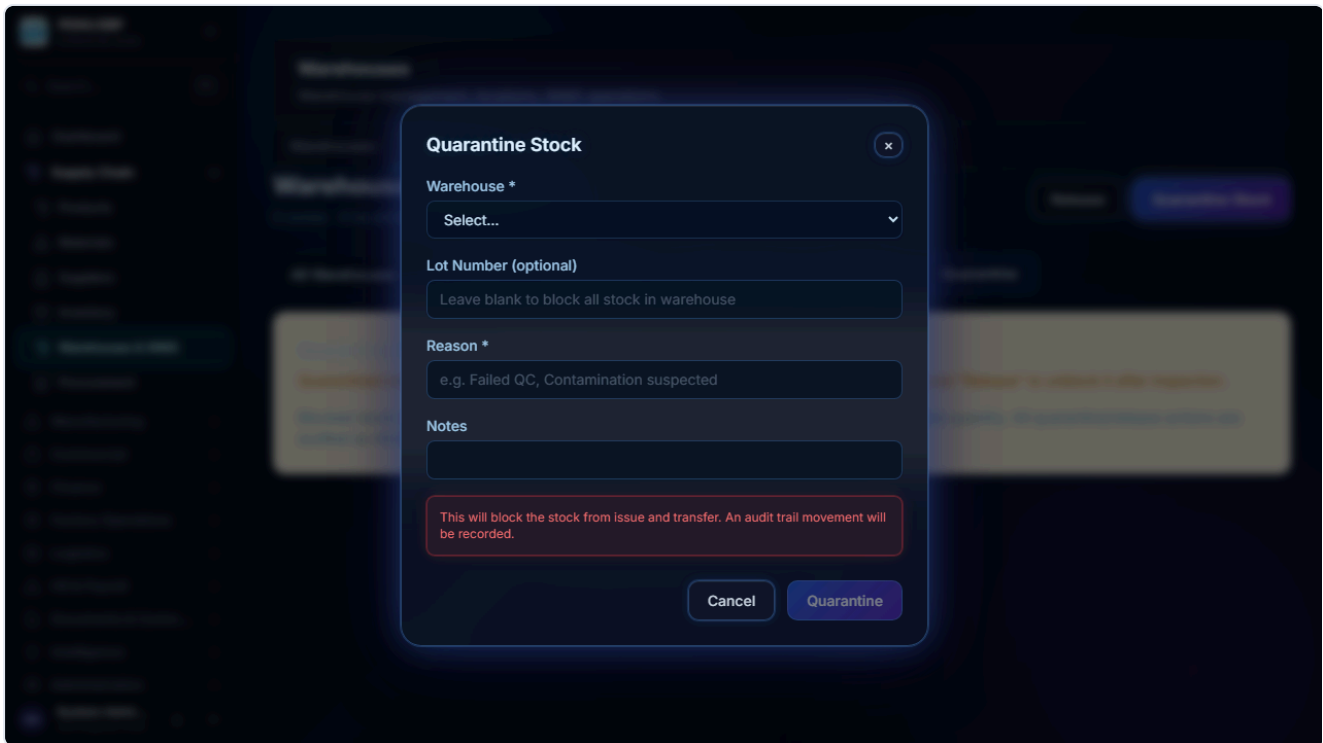
Wave status uses the same badge colour scheme as handling units.

Quarantine Tab

Tab key: `quarantine`



Quarantine tab showing quarantine action and release forms for blocking/unblocking lot stock.



Quarantine Stock modal — Warehouse, Lot Number, Reason, and Notes fields.

Quarantine blocks specific stock from being picked or issued. It applies at the lot level within a warehouse.

Quarantine Action Form

Field	Label	Required	Notes
<code>warehouse_id</code>	Warehouse	Yes	Warehouse containing the stock
<code>lot_number</code>	Lot Number	No	Leave blank to quarantine all lots in the warehouse
<code>reason</code>	Reason	Yes	Why the stock is being quarantined
<code>notes</code>	Notes	No	Additional detail

Effect: Sets `quantity_available = 0` for the affected stock positions. Creates an audit movement of type `QUARANTINE`.

Quarantine Release Form

Field	Label	Required	Notes
<code>warehouse_id</code>	Warehouse	Yes	Warehouse with quarantined stock
<code>lot_number</code>	Lot Number	Yes	Lot number to release

Field	Label	Required	Notes
notes	Notes	No	Release justification

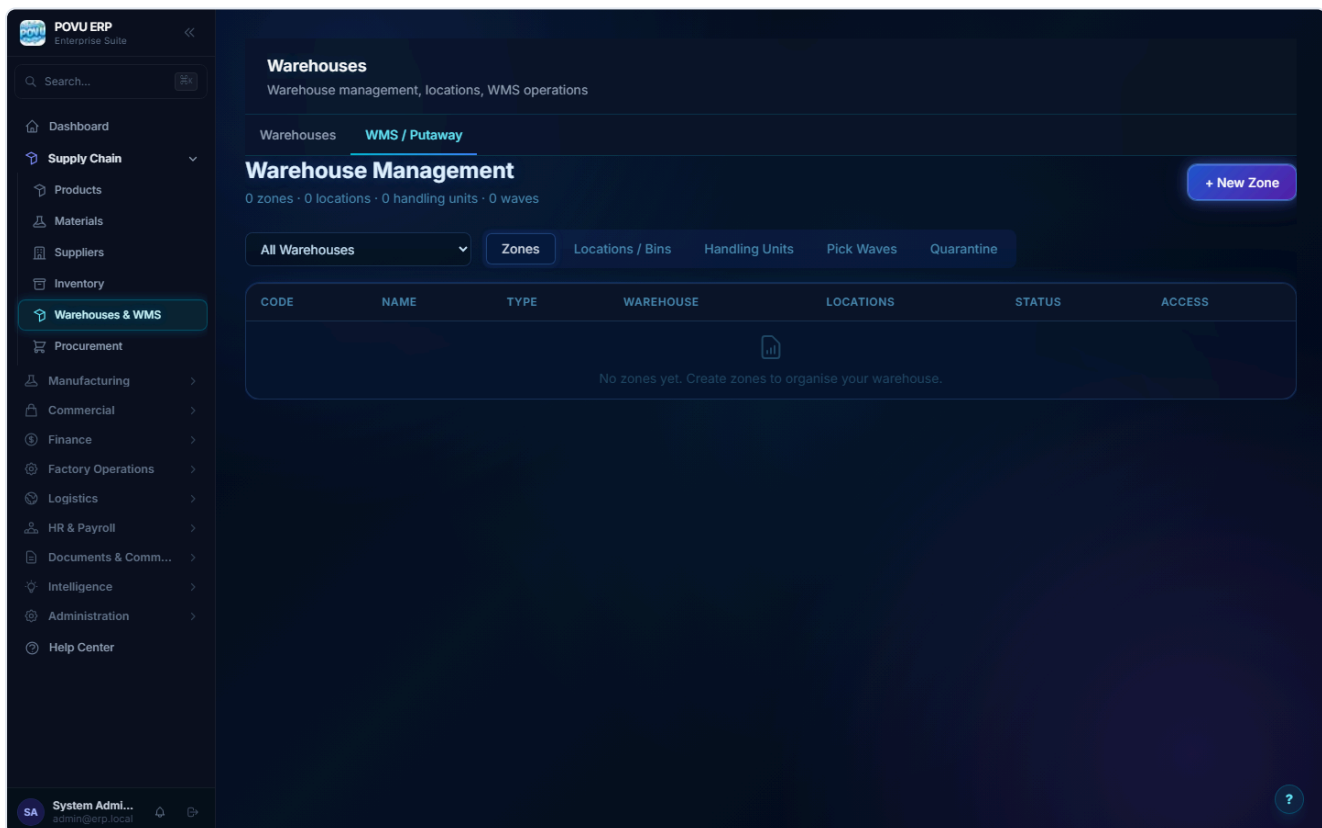
Effect: Restores `quantity_available` to prior level. Creates a `QUARANTINE_RELEASE` movement in the inventory ledger.

WMS and Inventory Relationship

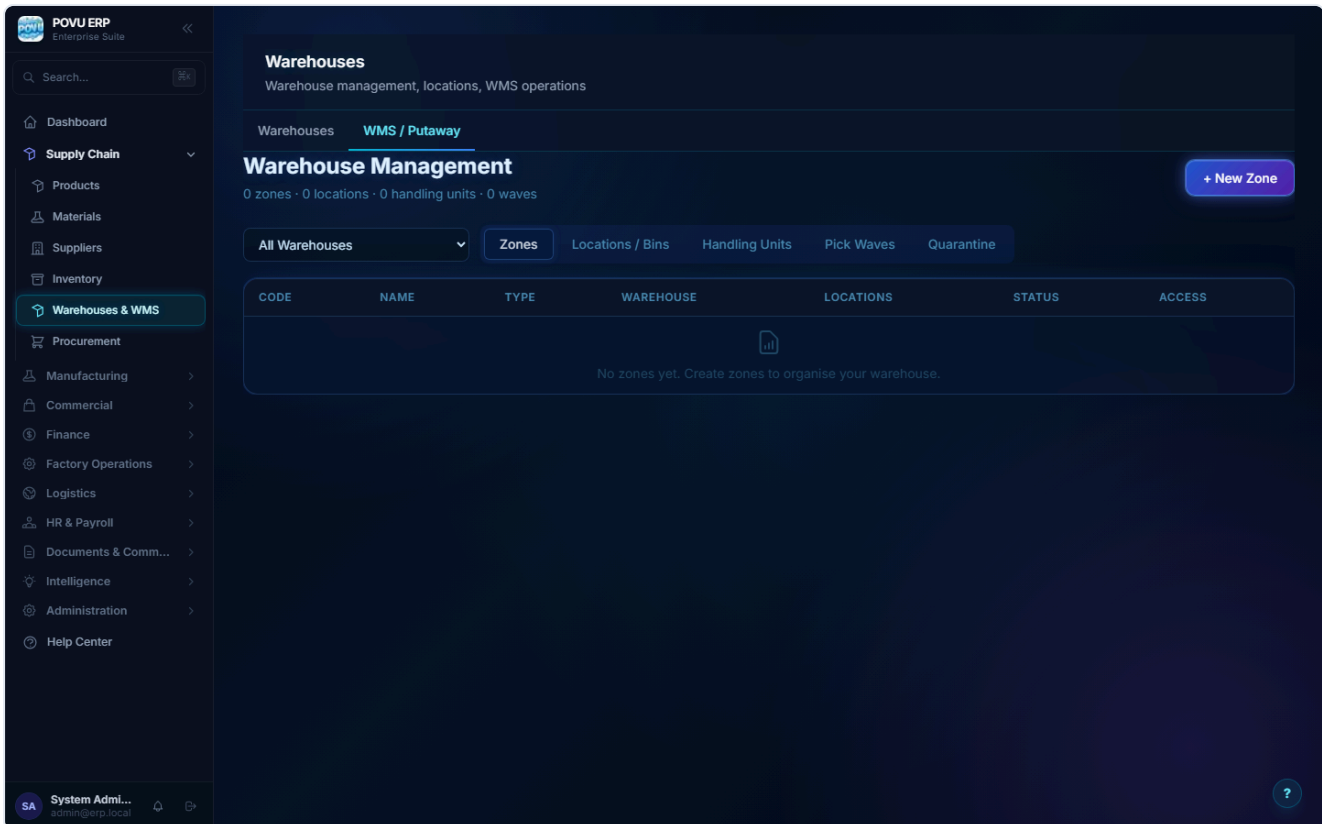
WMS locations are informational overlays on the inventory ledger. The authoritative stock quantity is always in the inventory ledger (`quantity_on_hand`). WMS provides the physical location data. Discrepancies between WMS and the ledger are resolved via cycle count.

Demo Data — WMS Seed (I2)

WMS zones and storage locations are seeded as part of the inventory demo data (TASK-016 I2). Both PROD-WH and FG-WH warehouses have pre-configured zones and bin locations.



WMS workspace showing seeded zones across PROD-WH and FG-WH warehouses.



WMS Locations tab with seeded bin locations including zone assignment and barcode.

Seeded Zone Types

Warehouse	Zone types seeded
PROD-WH	RAW_MATERIAL, SEMI_FINISHED, STAGING
FG-WH	FINISHED_GOODS, QUARANTINE, RETURNS

Seeded Bin Locations

Each zone has bin locations in aisle-rack-level format (e.g. `A-01-01` , `B-02-03`). Locations have:

- Zone assignment
- Scannable barcode (`LOC-{code}`)
- Active status
- Blocked status (false by default)

In a live deployment, replace seeded locations with the actual warehouse layout codes and barcodes before going live.

Logistics & Import Shipments

Route: /dashboard/logistics

Permission required: procurement.view

Workspace tabs: Overview, Shipments, Containers, Arrivals, Documents, Fleet

What It Does

The Logistics module manages international import shipments from origin port to warehouse delivery. It tracks vessels, containers, customs clearance, and arrival milestones. Logistics integrates with Procurement (links to POs) and Inventory (GRN creation on delivery).

The screenshot displays the POVU ERP Enterprise Suite interface. The left sidebar contains a navigation menu with options like Dashboard, Supply Chain, Manufacturing, Commercial, Finance, Factory Operations, Logistics (selected), HR & Payroll, Documents & Comm..., Intelligence, and Administration. The main content area is titled 'Logistics' and 'International Logistics', showing a supply chain visibility from Turkey to Kenya. It features four KPI tiles: 'ACTIVE SHIPMENTS' (0 in planning or transit), 'IN TRANSIT' (0 vessels at sea), 'AT PORT / CLEARING' (0 Mombasa port / customs), and 'OVERDUE DOCUMENTS' (0 draft or rejected docs). Below these is a 'Shipment Pipeline — Turkey to Kenya' table with columns for SHIPMENT, MODE, VESSEL / CARRIER, B/L, ORIGIN PORT, ETA, ETA (REVISED), DAYS, CTRS, POS, CLEARANCE, and STAT. The table currently shows 'No active shipments'. At the bottom, there are four action tiles: 'Shipment Planning' (Create and manage import shipments), 'Container Tracking' (Track container status and seal numbers), 'Customs Documents' (Invoice, BL, packing list, certificates), and 'Arrival & Clearance' (ETA alerts, port events, KRA clearance).

Logistics overview showing KPI tiles (Active Shipments, In Transit, At Port/Clearing, Overdue Documents) and ETA alert panel.

Dashboard KPIs

The overview tab shows four KPI tiles:

Title	Description
Active Shipments	Total shipments not cancelled or delivered
In Transit	Shipments with status <code>IN_TRANSIT</code>
At Port / Clearing	Shipments at <code>ARRIVED_PORT</code> , <code>CUSTOMS_HOLD</code> , or <code>CUSTOMS_CLEARED</code>
Overdue Documents	Count of shipments with overdue required documents; shown in red if > 0

ETA Alert Panel

Shipments within 14 days of ETA appear in the alert panel. Colour coding:

Condition	Display
<code>days_to_eta ≤ 3</code>	Red bold — critical; action required
<code>days_to_eta 4-14</code>	Amber — approaching

Shipment Status Values

Status	Colour	Meaning
<code>PLANNED</code>	—	Shipment being organised
<code>BOOKED</code>	—	Vessel / cargo space confirmed
<code>CARGO_LOADED</code>	Blue	Container loaded at origin port
<code>IN_TRANSIT</code>	Blue	On the vessel at sea
<code>ARRIVED_PORT</code>	Yellow	Arrived at destination port
<code>CUSTOMS_HOLD</code>	Red	Held by customs for inspection
<code>CUSTOMS_CLEARED</code>	Green	Customs clearance obtained
<code>OUT_FOR_DELIVERY</code>	Yellow	In transit from port to warehouse
<code>DELIVERED</code>	Green	Received at warehouse
<code>CANCELLED</code>	Gray	Shipment cancelled

Customs Clearance Status

Status	Colour
CLEARED	Green
HELD	Red
DUTY_PAID	Yellow

Transport Mode

Mode	Badge Colour
SEA	Blue
AIR	Yellow
ROAD	Gray

Shipments Tab

Tab key: `shipments`

POVM ERP Enterprise Suite

Search...

Dashboard

Supply Chain

Manufacturing

Commercial

Finance

Factory Operations

Logistics

HR & Payroll

Documents & Comm...

Intelligence

Administration

System Admin... admin@erp.local

Logistics
International shipments, containers, fleet, customs clearance

Overview **Shipments** Containers Arrivals Documents Fleet

Shipment Planning
Turkey → Kenya international shipments

New Shipment

ALL (0) DRAFT (0) BOOKING CONFIRMED (0) CARGO LOADED (0) IN TRANSIT (0) ARRIVED PORT (0) CUSTOMS HOLD (0)

CUSTOMS CLEARED (0) OUT FOR DELIVERY (0) DELIVERED (0)

SHIPMENT NO	MODE	ORIGIN	CARRIER / VESSEL	B/L	DEPARTURE	ETA	INCOTERM	CTRS	POS	STATUS
No shipments										

Shipments tab showing import shipments with mode badges, ETA, days to ETA (colour-coded), clearance status, and shipment status.

Lists all import shipments with filterable columns.

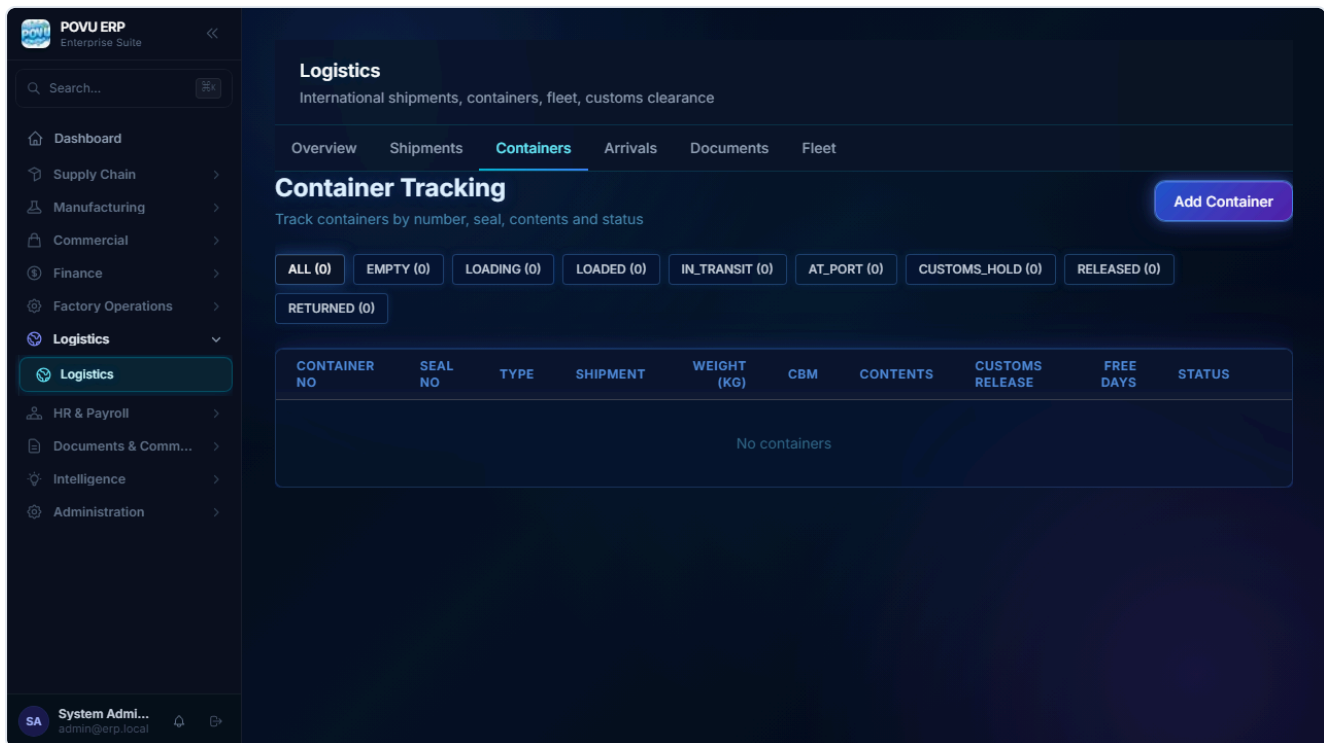
Shipment List Columns

Column	Field	Notes
Shipment No	<code>shipment_no</code>	Reference number
Mode	Mode badge	SEA / AIR / ROAD
Vessel / Carrier	<code>vessel_name</code> or <code>carrier</code>	Shipping line or air carrier
B/L Number	<code>bl_number</code>	Bill of lading
Origin Port	<code>origin_port</code>	Port of loading
ETA	<code>eta</code>	Original estimated time of arrival
ETA (Revised)	<code>eta_revised</code>	Updated ETA if rescheduled
Days to ETA	<code>days_to_eta</code>	Colour-coded by urgency
Containers	<code>container_count</code>	Number of containers on shipment
POs	<code>po_count</code>	Number of linked Purchase Orders
Clearance	<code>clearance_status</code>	Customs clearance badge
Status	<code>status</code>	Shipment status badge

Row highlight: Rows where `has_overdue_docs = true` display with a red-50 background.

Containers Tab

Tab key: `containers`

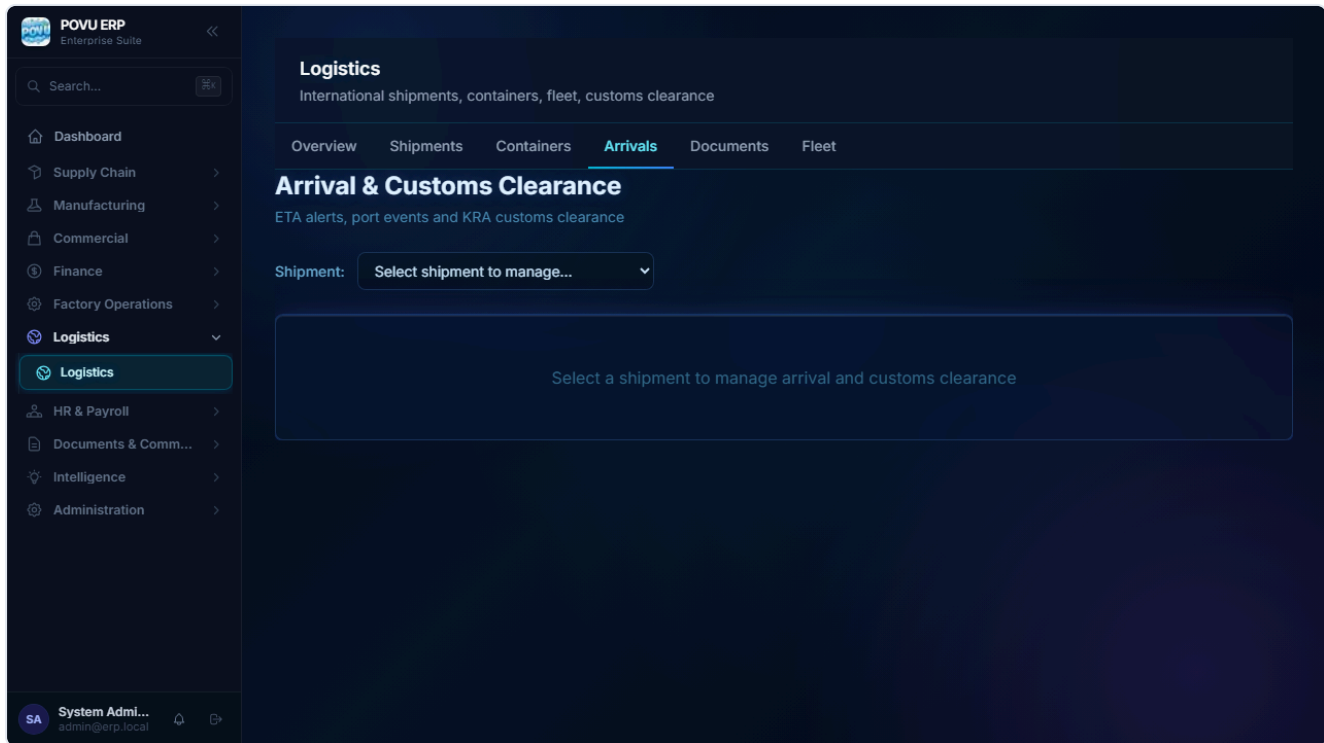


Containers tab showing container numbers, seal numbers, size, weight, and shipment linkage.

Tracks individual container records linked to shipments. Containers have their own tracking (seal number, container number, size, weight).

Arrivals Tab

Tab key: `arrivals`



Arrivals tab showing port event records, ETA alerts, and customs clearance tracking.

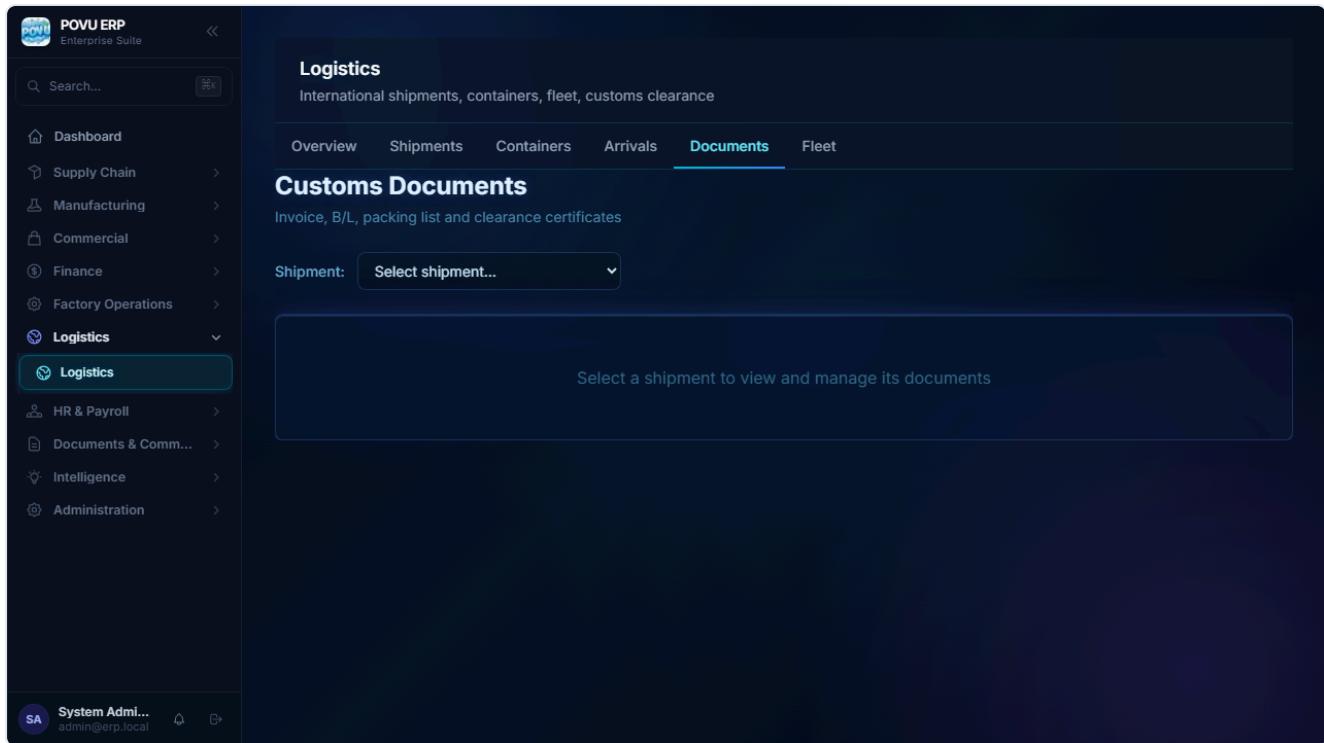
Records the physical arrival of goods at the warehouse. After an arrival is confirmed:

1. The logistics team marks the shipment as **DELIVERED**
2. The warehouse team creates GRNs (Goods Receipt Notes) in the Procurement module to credit inventory

See [04-deliveries.md](#) for the GRN process.

Documents Tab

Tab key: **documents**

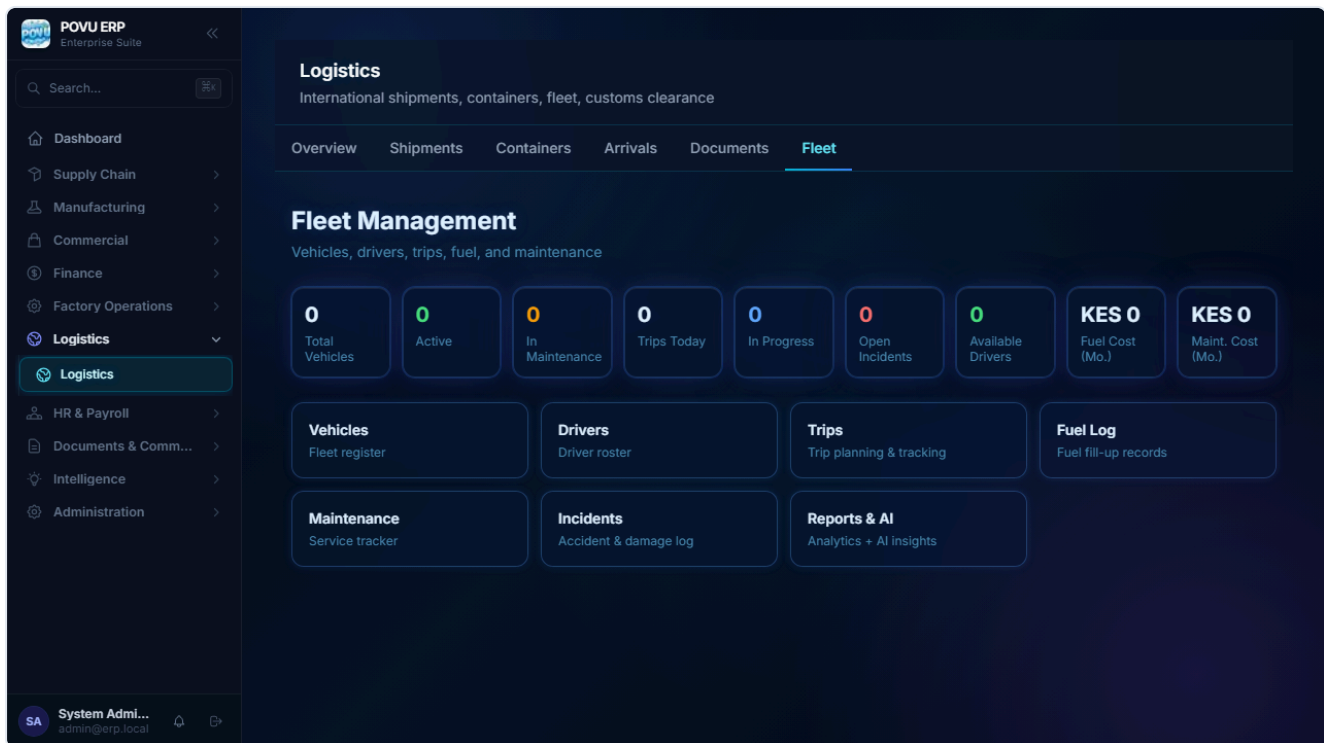


Documents tab listing shipping documents (bill of lading, commercial invoice, certificates) with due dates and status.

Manages import documentation: bill of lading, commercial invoice, packing list, certificate of origin, phytosanitary certificate, import declaration. Documents have due dates; overdue documents drive the red KPI tile on the dashboard.

Fleet Tab

Tab key: fleet



Fleet tab showing internal delivery vehicles with registration, driver assignment, and trip status.

Manages internal delivery fleet vehicles used for last-mile delivery from port to warehouse. Tracks vehicle registration, driver assignment, and trip logs.

Quick Links (Overview Tab)

Link	Purpose
Shipment Planning	Create and plan new import shipments
Container Tracking	Monitor container locations and status
Customs Documents	Upload and track clearance documents
Arrival & Clearance	Record arrivals and update clearance status

Logistics → Inventory Flow

```

Shipment BOOKED
  → Cargo loaded (CARGO_LOADED)
  → Vessel at sea (IN_TRANSIT)
  → Port arrival (ARRIVED_PORT)
  → Customs clearance (CUSTOMS_CLEARED)
  → Out for delivery (OUT_FOR_DELIVERY)

```

- Delivered to warehouse (DELIVERED)
- GRN created in Procurement
- Inventory RECEIPT movement posted
- Landed costs allocated to GRN lines